Visual Macroscopic Intelligence: A Survey from Multi-Image Understanding to Agentic Visual Reasoning

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Large image collections increasingly function as weakly organized visual contexts, where meaning depends on relations distributed across images. This survey develops *visual macroscopic intelligence* (VMI) as a framework for connecting image collection understanding, MLLM-based multi-image reasoning, visual RAG, and agentic visual evidence management. VMI treats collection understanding as a process of discovering latent structure, reasoning over cross-image evidence, and producing grounded artifacts such as summaries, timelines, reports, and memories. We review traditional image-sequence and photo-collection understanding, recent MLLM-based multi-image reasoning, latent structure induction, agentic visual reasoning, and datasets and benchmarks. The survey synthesizes an evidence-centered view of large visual contexts and identifies open challenges in scalable modeling, cross-image correspondence, faithful grounding, privacy-preserving memory, and real-world deployment.

CCS Concepts: • **Computing methodologies → Computer vision**; **Artificial intelligence**; *Natural language processing*; • **Information systems** → *Multimedia information systems*; *Information retrieval.*

Additional Key Words and Phrases: visual macroscopic intelligence, image collection understanding, multi-image understanding, agentic visual reasoning, multimodal large language models, multimodal agents

ACM Reference Format:
Zhiqiang Yuan, Zexi Jia, Ying Deng, Jiapei Zhang, Kailin Lv, Qiwei Yan, Yang Liu, Liwei Yang, Xiaoyue Duan, Peixiang Luo, Jinchao Zhang, and Jie Zhou. 2026. Visual Macroscopic Intelligence: A Survey from Multi-Image Understanding to Agentic Visual Reasoning. *ACM Comput. Surv.* 1, 1, Article 1 (May 2026), 57 pages.

1 Introduction

Many visual applications now operate on collections whose internal structure is only partially given. Personal photo albums, social-media galleries, shopping records, medical studies, remote-sensing

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observations, multi-view scene captures, and web documents all present images as groups of related, partially related, or weakly related observations. A phone album may interleave travel photos, receipts, screenshots, and repeated portraits; a domain repository may record the same object, patient, region, or document through different viewpoints, acquisition times, sensors, or page layouts. In such collections, useful information often appears through cross-image comparison: recurring entities reveal routines, timestamps expose event boundaries, repeated observations indicate change, and text-heavy screenshots or close-up views disambiguate scenes. These collections are valuable precisely because their semantics are distributed, noisy, and sometimes only recoverable after grouping and comparison. This survey asks how systems can turn such distributed visual observations into macroscopic structure, evidence, and usable knowledge. Figure 1 summarizes the setting.

Single-image and video paradigms cover only part of this setting. Single-image tasks operate inside one visual scene, while video understanding usually assumes dense temporal continuity. The collections considered here may be ordered, unordered, or weakly structured; they may contain sparse temporal cues, noisy metadata, near duplicates, missing intermediate states, and unrelated distractors. Reasoning over them requires a constructed visual context: the relevant units, relations, and evidence often have to be inferred before a system can answer, summarize, or generate. To make the regime concrete, we locate tasks along two practical dimensions: the scale and organization of the visual context, and the degree to which the goal, relevant subset, and expected output are specified in advance. A conventional multi-image question-answering task usually sits near the low-scale, high-specification end, because a few images and a question are already given. The settings considered in this survey extend to cases where the relevant visual units, relations, and evidence must first be inferred. We use *visual macroscopic intelligence* to denote methods that interpret large visual contexts by discovering latent structures, reasoning over cross-image evidence, and supporting downstream tasks such as selection, summarization, storytelling, question answering, grounding, comparison, event discovery, and report generation. In this survey, these tasks are treated as goal-conditioned outputs over a shared latent macroscopic structure: given a large visual context and an explicit or implicit goal, a system infers entities, events, temporal-spatial organization, relations, evidence links, and reusable knowledge before delivering a faithful answer, summary, timeline, report, or memory.

Early research already recognized the need to organize and summarize photo collections. Systems for personal albums and social photo storytelling explored how to select representative, important, and socially engaging images from user albums [1, 2], while visual storytelling introduced sequential vision-to-language generation and provided a large-scale dataset for mapping photo streams to coherent narratives [3]. These works established several enduring themes: image collections are redundant and uneven in quality; understanding them requires diversity, saliency, chronology, and narrative coherence; and the output is often a structured description, story, or summary. Traditional systems, however, were usually designed for specific tasks and relied on restricted pipelines, handcrafted signals, or task-specific supervision, limiting their ability to support open-ended reasoning, evidence synthesis, and macroscopic knowledge construction.

Recent multimodal large language models (MLLMs) have expanded multi-image understanding from a specialized setting into a common interface for interleaved image-text contexts, multi-image instruction following, multi-view perception, and visual reasoning across images. Representative models and training frameworks such as MANTIS and LLaVA-NeXT-Interleave show that instruction tuning with interleaved multi-image data can improve co-reference, comparison, temporal reasoning, and multi-image dialogue capabilities [4, 5]. Benchmarks such as BLINK, MuirBench, MMIU, and All-Angles Bench also show that strong MLLMs still struggle with precise spatial

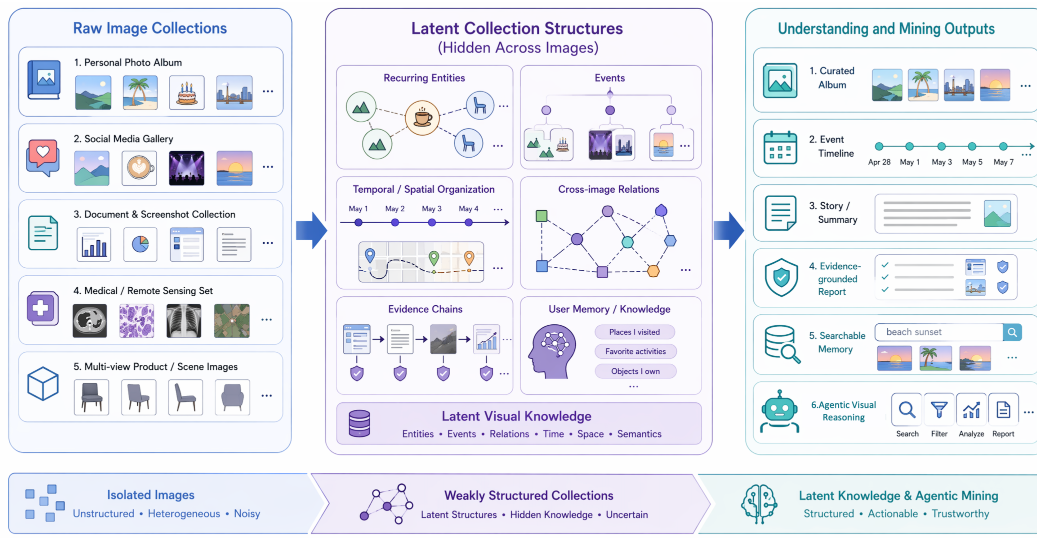

Fig. 1. Overview of visual macroscopic intelligence. Visual data appears as unordered or weakly structured collections rather than isolated images. The goal is to infer recurring entities, events, temporal-spatial patterns, and evidential relations at a macroscopic scale, then produce user-facing outputs such as curated albums, timelines, summaries, reports, searchable records, or evidence artifacts produced through agentic reasoning.

relations, cross-image comparison, temporal ordering, cross-view correspondence, and visual distractors [6–9]. Multi-image understanding has therefore become a research problem in its own right.

The neighboring literatures expose complementary pieces of the same problem. Photo-album summarization and visual storytelling treat images as collections, yet usually assume domain-specific pipelines and fixed output forms such as representative subsets or narratives. Multi-image MLLMs generalize visual-language reasoning to several images, but most benchmarks and models still operate on a small, preselected set of images with an explicitly given question. Visual-document RAG and multimodal retrieval provide mechanisms for accessing evidence, but they are typically query-centric and leave latent visual units, cross-image relations, and reusable knowledge outside the task definition. Multimodal agents contribute planning, tool use, and iterative verification, while existing agent surveys seldom take large weakly structured visual contexts as the object of interpretation. We use *visual macroscopic intelligence* to name the shared problem of constructing a task-relevant evidence state over a large visual context and producing grounded artifacts such as summaries, timelines, reports, answers, or memories.

VMI is broader than current multi-image MLLM evaluation because many benchmarks choose the unit of analysis for the model: the relevant images are provided, the question is specified, and the expected output is usually an answer or short explanation. Macroscopic visual settings remove some of these conveniences. A phone album may require the system to separate a trip from daily screenshots, group near-duplicate portraits, and recognize that a receipt image is relevant to a later expense query. A medical or remote-sensing archive may require comparing views or time points before deciding which change deserves reporting. These examples put context construction inside the task itself.

Agentic systems offer one operational pattern for this setting. Language agents with reasoning-and-acting loops can decompose tasks, invoke APIs, and revise plans based on observations [10, 11]. When paired with multimodal perception and visual-document RAG systems such as VisRAG [12], an agent can sample a collection, run OCR on text-heavy images, compare candidate duplicates,

Table 1. Positioning visual macroscopic intelligence relative to neighboring areas. VMI is framed as the regime where latent structure, evidence, and macroscopic artifacts must be constructed over large visual contexts.

Area	Main focus	Task form	VMI gap	Role in this survey
Image and composed retrieval [13, 14]	Indexing, ranking, and image-text query composition.	Query → ranked images.	Treats the collection mainly as a search space, not as structure to be interpreted.	Evidence access, deduplication, sub-context selection, and scalable exploration.
MLLM multi-image understanding [4, 5, 7, 8]	Multi-image QA, comparison, and instruction following.	Few images → answer.	Usually assumes preselected images and a specified question.	Low-scale, high-specification case within the VMI space.
Visual-document RAG and document QA [12, 17]	Retrieving visual or document evidence for QA and generation.	Corpus + query → answer.	Supports evidence access, but not latent visual-unit or artifact induction.	Substrate for evidence-state construction, verification, and faithful generation.
Text reasoning-intensive retrieval [15]	Retrieving textual evidence requiring nontrivial reasoning.	Text query → documents.	Omits visual evidence, cross-image correspondence, and weak visual structure.	Textual analogue for reasoning-aware evidence access.
Multimodal agents [16]	Planning, tool use, interaction, and memory.	Task + tools → traces.	Does not center large visual contexts, provenance, or evidence states.	Agentic visual reasoning is treated as a VMI system pattern.
Photo collection and storytelling [1, 3, 18]	Album organization, selection, event grouping, and narratives.	Album/stream → story.	Models collections within fixed outputs and limited provenance constraints.	Traditional primitives for structure induction, evidence selection, and artifacts.

or invoke a detector for ambiguous cases. Selective inspection becomes useful when exhaustive visual-context processing is infeasible, and it motivates agentic visual reasoning as a system pattern for evidence-state construction in VMI.

The survey builds on adjacent areas while focusing on the layer they leave underdeveloped. CSUR's image-retrieval synthesis shows how visual collections become searchable resources [13], composed-image retrieval surveys cover image-text query composition [14], reasoning-intensive retrieval benchmarks such as BRIGHT expose multi-step evidence access in text [15], and multimodal-agent surveys summarize planning and tool-use mechanisms [16]. VMI asks how visual collections become latent structures, evidence states, and grounded artifacts, not only ranked lists or single answers. Table 1 compares these neighboring areas with the coverage of VMI.

Work on image-collection analysis, MLLM-based multi-image reasoning, visual RAG, long-context multimodal modeling, and multimodal agents has expanded rapidly since 2023, yet the results remain split across separate vocabularies, assumptions, and benchmarks. This fragmentation makes it difficult to compare capabilities or to see which parts of large visual-context reasoning are still unevaluated. The survey addresses the gap with an orthogonal taxonomy and a latent evidence-graph abstraction that connect structure discovery, evidence grounding, agentic reasoning, and benchmark design.

Figure 2 organizes the paper around a progression from multi-image understanding to agentic visual reasoning. We connect visual storytelling, album summarization, multi-image question answering, grounding, and agent-based analysis through one operational question: what capabilities are needed when the useful units of a large visual context must be inferred? Early photo-collection research, recent MLLM-based reasoning, and emerging agentic systems all touch this question,

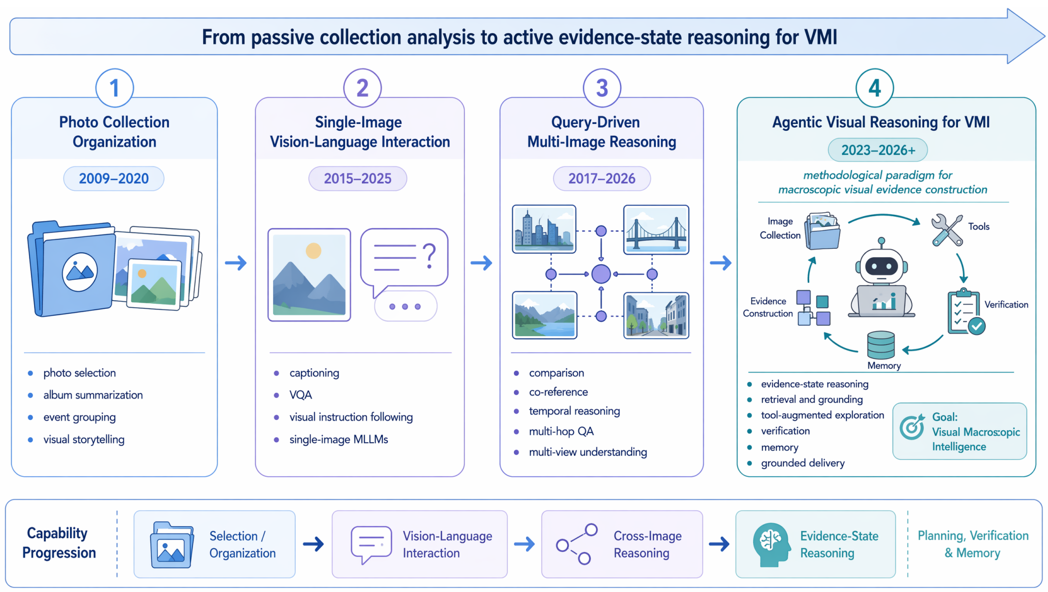

Fig. 2. Capability evolution toward agentic visual reasoning for visual macroscopic intelligence. The figure summarizes a representative progression from photo collection organization and single-image vision-language interaction to query-driven multi-image reasoning and agentic visual reasoning for VMI. The progression reflects a shift from passive collection analysis to active evidence-state reasoning, where planning, tool use, verification, and memory support the construction of macroscopic visual evidence and grounded artifacts.

but usually under different names. Existing labels often emphasize an input format, a retrieval mechanism, an application domain, or an agent architecture; emerging applications require these components to be composed around a shared intermediate object, namely a macroscopic visual evidence state. VMI serves here as an organizing lens for weakly structured visual big data, with latent structure induction, evidence-constrained generation, and agentic exploration treated as coupled parts of the same problem.

The main contributions of this survey are as follows:

- We formulate *visual macroscopic intelligence* as a paradigm for interpreting large visual contexts, unifying selection, summarization, question answering, retrieval, reports, and memory as goal-conditioned transformations from latent structure to grounded artifacts.
- We synthesize a latent evidence-graph view over entities, events, relations, claims, provenance, and memory, clarifying how VMI differs from single-image understanding, video understanding, and query-driven multi-image reasoning.
- We organize traditional photo-collection analysis and recent MLLM-based multi-image understanding within the same taxonomy, showing how selection, storytelling, grounding, long-context modeling, and benchmark design correspond to different stages of VMI.
- We position *agentic visual reasoning* as a system pattern for VMI, and identify open challenges in scalability, correspondence, faithfulness, privacy, and evaluation.

2 Survey Scope and Literature Search Methodology

Scope and objective. This survey is designed as a structured scoping survey rather than an exhaustive systematic review, consistent with the role of scoping and mapping reviews in clarifying concepts, boundaries, and evidence landscapes for emerging fields [19–21]. We discuss adjacent fields selectively when they contribute to macroscopic visual organization, reasoning, evidence attribution, or grounded knowledge construction. The paper set was constructed through multi-stage keyword search, backward reference tracing, and forward citation tracing over work published or released up to May 2026. To make the coverage process auditable, Table 2 summarizes the simplified scoping flow from identified records to the full coded corpus, while the supplementary material reports the full search protocol, boundary criteria, orthogonal taxonomy, quantitative publication landscape, and survey limitations.

Search strategy. Candidate papers were collected from arXiv, Google Scholar, ACM Digital Library, IEEE Xplore, CVF Open Access, ACL Anthology, and DBLP. The seed queries covered six families: traditional collection analysis and retrieval; MLLM-based multi-image understanding; relation, grounding, and correspondence; long-context and visual-RAG settings; spatial and multi-view foundations; and agentic mechanisms. Keyword search was used as a seed rather than as a complete census, because relevant work is often indexed under neighboring terms such as visual storytelling, visual correspondence, document-image reasoning, visual RAG, or multimodal agents. Backward tracing added older collection-organization, retrieval, storytelling, and multi-view foundations; forward tracing from datasets, benchmarks, and representative systems added recent multi-image, grounding, and visual-RAG work that uses rapidly changing terminology.

Screening and coding. We retained papers when they satisfied at least one of four criteria: treating a sequence, set, or repository of images as the main input; producing a macroscopic output such as a representative subset, event grouping, story, report, grounded answer, or benchmark label; studying MLLMs under multi-image, long-context, grounding, or evidence-search settings; or introducing retrieval, tool-use, state, memory, or agent mechanisms that can support collection-level visual analysis. We excluded pure single-image recognition, captioning, or VQA unless they serve as foundations for multi-image reasoning; dense video understanding unless it informs sparse image-sequence or collection-level reasoning; generic image retrieval without evidence construction or organization; and general-purpose agents without explicit visual evidence handling. Each included work was then coded by publication year, dominant role, input structure, primary capability, evaluation setting, and VMI relevance.

Limitations. This scoping design also introduces several limitations. The corpus is intended to map concepts, task formulations, capabilities, and evaluation gaps rather than to provide a complete bibliometric census of image retrieval, video understanding, MLLMs, visual RAG, or multimodal agents. The boundary remains unstable because many relevant 2025–2026 works are preprints, technical reports, or rapidly updated benchmark releases. Benchmark statistics are also not directly comparable across papers, since different resources report images, frames, pages, slides, questions, dialogues, visual histories, or retrieval units under different conventions. We therefore use these numbers mainly to characterize input scale and evaluation scope, while relying on explicit inclusion criteria and capability coding to handle boundary cases.

Table 2. Compact scoping flow for the full coded literature corpus, including works discussed in the main text and supplementary material. Counts refer to retained candidate records rather than raw platform hit counts.

Stage	Records	Main action
Identified	629	Keyword search plus backward/forward tracing.
Deduplicated	487	Merge by title, DOI, arXiv ID, and venue version.
Title/abstract screened	487	Exclude off-scope single-image, dense-video, generic-retrieval, and non-visual-agent work.
Full text assessed	285	Check VMI relevance, boundary role, and contribution type.
Included and coded	205	Final corpus for taxonomy and publication-trend coding.

3 Background and Problem Definition

This section formalizes the scope of visual macroscopic intelligence and clarifies its relationship to adjacent visual understanding paradigms. The goal is to establish a common vocabulary for the subsequent discussions of traditional photo-collection analysis, MLLM-based multi-image understanding, latent structure induction, and agentic visual reasoning.

3.1 Definition and Scope of Visual Macroscopic Intelligence

We define a *large visual context* as a set of images $C = \{I_1, I_2, \ldots, I_N\}$, where each image I_i may optionally be associated with metadata m_i, such as timestamp, location, device information, filename, user tags, textual content, or album membership. Unlike a single image, C provides a macroscopic visual context: useful information may be distributed across images and may only become evident after comparing, grouping, ordering, or aggregating multiple visual observations.

Visual macroscopic intelligence refers to the problem of interpreting such a context by discovering visual content, latent structures, evidential relations, and task-relevant knowledge. Image collection understanding is the core problem setting within this broader framing. This view is related to early image-mining studies, which emphasized extracting implicit patterns, semantic information, and application-level knowledge from large image repositories [22], but it further foregrounds multimodal reasoning, evidence grounding, and agentic control. The boundary is best understood as a two-dimensional continuum rather than a single threshold. Along the *scale axis*, VMI ranges from small curated image sets to large, heterogeneous repositories; along the *task-specification axis*, it ranges from explicit questions to exploratory goals for which the relevant subset, latent structure, and output artifact must first be discovered. Query-driven multi-image understanding is included as a restricted case with a small or preselected visual context and a specified goal, whereas full VMI additionally requires collection organization, evidence selection, and structure discovery before answering or generating.

Operationally, a task moves toward the VMI regime when at least one assumption of standard multi-image understanding is relaxed: the relevant image subset is not fully specified, the relations among images must be inferred, the output requires evidence attribution across images, or the result is intended to serve as a reusable artifact rather than a one-shot answer. Conversely, a two-image comparison question with preselected inputs is better viewed as conventional multi-image understanding, and a query that only retrieves visually relevant pages without inducing cross-image structure is better viewed as visual RAG. VMI therefore does not replace these areas; it describes the regime in which their components must be composed to construct macroscopic visual knowledge. Formally, a system is expected to infer a structured representation $\mathcal{Z} = \{\mathcal{E}, \mathcal{R}, \mathcal{T}, \mathcal{S}, \mathcal{K}\}$, where $\mathcal{E}$ denotes entities or events, $\mathcal{R}$ denotes cross-image relations, $\mathcal{T}$ denotes temporal or spatial organization, $\mathcal{S}$ denotes summaries, stories, or reports, and $\mathcal{K}$ denotes reusable knowledge or memory. The formulation emphasizes outputs that depend on relations among images, such as an event timeline, a before-after comparison, or a report with cited visual support.

3.2 A Latent Evidence-Graph View

To make the above definition operational, we use a latent evidence-graph view as an organizing abstraction rather than propose it as a new model. The abstraction is distilled from two complementary traditions: knowledge-graph and provenance models, which represent entities, relations, claims, and support records explicitly [23–25], and visual grounding resources, which link language, objects, regions, and relations in images [26–28]. Let

$$G_C = (V, E, A),$$

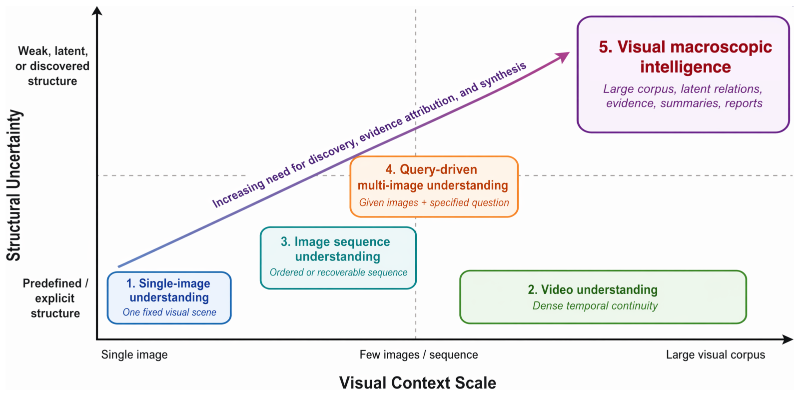

Fig. 3. Conceptual positioning of visual macroscopic intelligence among adjacent visual understanding paradigms. The horizontal axis denotes increasing visual context scale, from a single image to a large visual context, while the vertical axis denotes structural uncertainty, from predefined or explicit structure to weak, latent, or discovered structure. Visual macroscopic intelligence occupies the high-scale, high-uncertainty region, where entities, events, relations, evidence, and outputs must be discovered jointly.

where V contains image nodes, entity nodes, event nodes, claim nodes, and memory nodes; E contains relations such as *same-as*, *part-of*, *before/after*, *supports*, *contradicts*, and *complements*; and A stores attributes such as timestamps, locations, OCR snippets, captions, confidence estimates, and provenance. Under this view, album summarization, visual storytelling, multi-image question answering, visual-document RAG, and agentic visual reasoning differ mainly in which subgraph is queried, which output artifact is required, and how strict the evidence constraints are. The supplementary material provides a concrete illustrative instantiation of such an evidence graph for a weakly structured personal image collection.

This graph view yields a unified task formulation. Given a large visual context C and an explicit or implicit goal g, the system first induces a task-relevant latent structure and then produces an output:

$$Z^* = \arg\max_Z p(Z \mid C, g), \qquad Y^* = \arg\max_Y U(Y; g, Z).$$

Here Z is a structured hypothesis about entities, events, relations, temporal-spatial organization, evidence, and memory, while Y may be a curated subset, answer, story, timeline, report, index, or memory update. Faithful macroscopic generation additionally requires an evidence constraint:

$$\forall c \in \text{Claims}(Y),\ \exists E_c \subseteq C : \text{Support}(E_c, c) > \tau.$$

Each factual claim in the output should therefore be supported by sufficient image-level, region-level, metadata-level, OCR-level, or tool-derived evidence. We use this graph view in three ways: to treat diverse applications as different goals over C, to represent intermediate products as components of G_C, and to compare retrieval pipelines, MLLMs, visual RAG, and agents as mechanisms for inducing, querying, updating, and verifying the latent evidence graph.

3.3 Relation to Existing Visual Understanding Paradigms

Visual macroscopic intelligence is related to, but distinct from, several established paradigms in computer vision and multimodal learning. Figure 3 provides a conceptual positioning of these paradigms along two axes: the scale of visual context and the degree to which the input structure is predefined or latent.

Single-image understanding focuses on one visual scene and has driven progress in object recognition, captioning, visual question answering, grounding, and image-text pretraining, as exemplified by large-scale recognition and VQA benchmarks and by region- or transformer-based vision-language models [29–32]. However, it cannot represent information that emerges only across multiple images, such as whether two photos depict the same person, whether an object changes over time, or which subset of an album best summarizes an event.

Video understanding processes dense temporal signals and is naturally suited to action recognition, event detection, and video question answering, with spatio-temporal action recognition being a representative direction [33]. Large visual contexts, however, are usually sparse and irregular. A personal album may contain only a few photos from an event, omit important intermediate states, include unrelated distractors, or mix images from different days and locations. Therefore, VMI cannot simply be reduced to video understanding.

Image sequence understanding assumes an ordered or recoverable sequence of images. Visual storytelling is a representative task in which systems generate coherent narratives from photo streams [3]. This paradigm captures temporal and narrative aspects of image collections but often presumes a relatively clean sequence and a predefined generative objective.

Multi-view geometry and robotic perception study image collections whose cross-image structure is governed by camera motion, scene geometry, and spatial consistency. Classical multiple-view geometry, structure-from-motion, photo-tourism systems, and visual SLAM establish how correspondence, pose, triangulation, and map consistency can be recovered from overlapping views [34–37]. This literature is not usually framed as visual macroscopic intelligence because its primary output is geometric reconstruction or localization rather than a user-facing semantic artifact. Nevertheless, it provides a crucial technical lineage for VMI: many cross-image correspondence and multi-view reasoning failures in MLLMs are precisely failures to preserve entity identity, viewpoint relations, and spatial evidence constraints that geometric vision has long treated explicitly.

Multi-image understanding has recently become prominent in the MLLM literature. Models such as MANTIS, LLaVA-NeXT-Interleave, LongLLaVA, and Qwen2-VL are trained or adapted to handle interleaved image-text inputs, long visual contexts, videos, or multiple images [4, 5, 38, 39], while benchmarks such as MuirBench, MMIU, MMNeedle, and Visual Haystacks evaluate multi-image relations, reasoning, robustness, and long-context retrieval [7, 8, 40, 41]. Spatial and multi-view benchmarks such as VSI-Bench and All-Angles Bench further target egocentric spatial intelligence, cross-view correspondence, and camera-pose reasoning [9, 42]. These studies provide essential foundations, but they are typically query-driven: the system receives a limited number of images and answers a specific question. In our terminology, this is not a competing definition of VMI but a restricted operating point within it. We nevertheless list query-driven multi-image understanding separately because it isolates a widely studied low-scale, high-specification setting, whereas visual macroscopic intelligence emphasizes the additional cases in which the relevant question, support, and internal grouping may not be specified in advance. Table 3 summarizes these differences across adjacent paradigms.

Agentic visual reasoning is therefore discussed later as a promising implementation strategy for visual macroscopic intelligence, where tools, memory, retrieval, and iterative verification help explore large visual contexts and produce grounded outputs.

3.4 Types of Image Collections

Image collections can be categorized by the structure available to the model.

Ordered image sequences. These collections provide an explicit order, such as story images, instructional steps, screenshots of a workflow, before-after pairs, or longitudinal observations. The main challenge is to model transitions, temporal dependencies, and narrative coherence.

Unordered image sets. These collections contain multiple images without a reliable order. Examples include personal albums, product galleries, retrieved image groups, social-media posts,

Table 3. Comparison of visual macroscopic intelligence with adjacent visual understanding paradigms. Query-driven multi-image understanding is treated as a restricted point within the broader VMI space, while agentic visual reasoning is treated as a system-level approach to VMI rather than as a separate peer paradigm.

Paradigm	Input	Structure	Primary goal	Boundary relative to visual macroscopic intelligence
Single-image understanding	One image	Fixed visual scene	Recognition, captioning, grounding, or VQA	Lacks cross-image relations, evidence, and distributed knowledge.
Video understanding	Dense frames or clips	Temporal order and continuity	Action, event, or temporal reasoning	Assumes dense continuity, unlike sparse or irregular visual contexts.
Image sequence understanding	Ordered image sequence	Given or recoverable order	Story generation, temporal ordering, or sequence-level summarization	Assumes a clean sequence and predefined narrative or temporal objective.
Multi-view geometry / robotic perception	Overlapping views or sensor streams	Geometry-, pose-, or map-constrained structure	Reconstruction, localization, mapping, or spatial perception	Provides correspondence tools but lacks semantic artifacts and evidence-aware synthesis.
Query-driven multi-image understanding	A few or preselected images	Query-specified input set	Answer a predefined question or instruction	Restricted low-scale, high-specification VMI case; bypasses discovery and evidence selection.
Visual macroscopic intelligence	Real-world large visual context, from small sets to large repositories	Latent, weak, or noisy cross-image structure	Discover macroscopic structure, evidence, and task-relevant knowledge	Adds latent structure induction, evidence attribution, sub-context discovery, and artifact generation.

and multi-view object images. The system must infer grouping, relevance, redundancy, and cross-image relations from visual content or weak metadata.

Weakly structured image collections. Many real collections lie between these two extremes. They may include timestamps, GPS traces, EXIF attributes, folder names, captions, or user tags, but these signals can be incomplete, noisy, or misaligned with the semantic structure of the collection. Weak structure is particularly common in user albums and domain repositories, making it a central setting for VMI.

3.5 Latent Structures in Image Collections

The value of an image collection often lies in latent structures that are not explicit in any single image. We highlight several recurring types.

Entity structure captures recurring people, objects, documents, landmarks, products, or visual motifs. It supports tasks such as de-duplication, entity tracking, and cross-image grounding.

Event structure groups images into meaningful activities or episodes, such as trips, meetings, meals, medical examinations, construction stages, or remote-sensing changes. Event structure provides the basis for album organization, summarization, and report generation.

Temporal and spatial structure describes ordering, progression, repeated routines, geographic trajectories, or multi-view spatial configurations. These structures are crucial for before-after reasoning, longitudinal analysis, and multi-view understanding.

Relational and evidential structure identifies how images support, contradict, complement, or contextualize one another. For example, one image may provide a close-up of an object, another may show its surrounding scene, and a third may contain text evidence. Such relations are central to faithful multi-image reasoning.

Narrative and user-level structure connects visual observations into stories, memories, preferences, habits, and long-term user context. This structure is especially important for personal photo assistants and agentic systems that maintain memory across multiple interactions.

3.6 Task Formulation and Expected Outputs

Visual macroscopic intelligence supports a broad family of tasks. At the organization level, systems may cluster images, recover temporal order, identify events, remove redundancy, or select representative images. At the reasoning level, they may answer questions, compare images, infer changes, ground evidence, or explain relationships. At the generation level, they may produce captions, stories, summaries, reports, timelines, or personalized memories. At the system-process level, they may actively retrieve relevant images, call external tools, update memory, and iteratively refine hypotheses through explicit reasoning traces.

The rest of the survey follows this hierarchy. Section 4 reviews traditional image sequence and photo collection methods. Section 5 summarizes MLLM-based multi-image understanding. Section 6 discusses latent structure induction for VMI, and Section 7 presents agentic visual reasoning as the system paradigm for VMI.

4 Traditional Image Sequence and Photo Collection Understanding

Before the emergence of MLLM-based multi-image reasoning, a long line of research had already studied how to select, organize, summarize, and narrate image sequences and photo collections. These works were often motivated by practical applications such as personal album management, slideshow generation, photobook authoring, social photo sharing, and visual storytelling. Although their goals were narrower than the open-ended setting considered in this survey, they established important primitives for visual macroscopic intelligence: selecting representative images, reducing redundancy, grouping photos into events, modeling temporal or narrative order, and generating compact user-facing descriptions.

The deeper historical root lies in image and multimedia data mining. Early image-mining work framed image repositories as sources of implicit patterns, semantic concepts, associations, and application-level knowledge rather than as isolated recognition inputs [22]. In parallel, content-based image retrieval and large-scale visual search developed representations and indexes for searching image collections by visual similarity, local invariant features, visual vocabularies, compact descriptors, and spatial verification [43–47]. These techniques did not solve open-ended macroscopic reasoning, but they supplied the substrate on which VMI still depends: feature extraction, similarity search, clustering, duplicate handling, scalable indexing, and candidate evidence construction. The difference is that VMI elevates these operations from collection-access mechanisms to components of a process that induces latent structure and generates grounded artifacts.

Representative traditional task families include mining and indexing, selection and curation, event organization, temporal ordering, narrative abstraction, and story-quality or grounding analysis. Detailed discussion of these task families and the traditional-task taxonomy is provided in the supplementary material. Here we focus on how these primitives carry forward into modern visual macroscopic intelligence.

4.1 From Traditional Primitives to Modern VMI Modules

Although traditional collection-analysis systems were often designed for narrow applications, many of their operations reappear as functional modules in modern visual macroscopic intelligence. Representative selection becomes evidence selection under visual-context budgets; album summarization becomes event-graph induction; redundancy removal becomes context compression and retrieval deduplication; temporal ordering becomes timeline construction and change reasoning; and visual

Table 4. From traditional collection-analysis primitives to modern visual macroscopic intelligence modules. Traditional methods are not simply replaced by MLLMs; many become reusable operations inside retrieval, indexing, evidence selection, structure induction, and verification loops.

Traditional primitive	Original role in collection analysis	Modern VMI counterpart	Technical continuity
Representative selection	Select a compact, diverse, high-quality subset for albums, slideshows, or photobooks [1, 2].	Evidence subset selection, retrieval reranking, context compression, and budget-aware inspection.	Coverage, diversity, quality, and importance constraints become mechanisms for deciding which visual evidence should enter an MLLM or agent context.
Redundancy removal and grouping	Reduce near duplicates and cluster visually or temporally related photos for concise browsing [1, 18].	Deduplication, cluster-aware retrieval, context indexing, and candidate-set compression.	Similarity and redundancy modeling become infrastructure for scalable macroscopic visual search and for avoiding duplicate-heavy evidence sets.
Event segmentation and album organization	Segment personal photos into events or episodes using timestamps, metadata, visual similarity, and social context [2, 48].	Latent event-graph induction, sub-context discovery, timeline construction, and task-specific grouping.	Event clusters become hypotheses about the latent structure Z that must be queried, revised, and verified for a user goal.
Temporal ordering and storyline reconstruction	Recover plausible temporal, causal, or narrative order from weakly ordered image streams [49, 50].	Before-after reasoning, state-transition analysis, change detection, and evidence-chain construction.	Ordering models provide the precursor to temporal edges in an evidence graph and to claims about progression or change.
Story generation and narrative planning	Compress a short sequence into coherent story-like language or procedural descriptions [3, 51].	Grounded reports, timelines, explanations, and user-facing macroscopic artifacts.	Narrative coherence remains useful, but modern VMI additionally requires claim-level provenance and uncertainty control.
Scene graphs and QA plans	Use objects, relations, scene graphs, or question-answer plans as intermediate structures for retrieval and story generation [52, 53].	Latent evidence graphs, claim planning, verification checklists, and structured reasoning state.	Object-relation structures are extended from local narrative support to macroscopic evidence, contradiction, and provenance modeling.
User and social preference modeling	Personalize selection or storytelling according to user, social, or event-specific importance [2, 18].	User-conditioned retrieval, persistent visual memory, preference-aware delivery, and privacy-sensitive personalization.	Preference signals become long-term context that must remain grounded, editable, and privacy-aware.

storytelling becomes grounded artifact generation. The main difference is not that these primitives disappear, but that they are embedded in open-ended, goal-conditioned, and evidence-constrained reasoning systems.

Table 4 summarizes this technical continuity. Traditional systems usually optimized a fixed output, such as a slideshow, album summary, or story. Modern VMI systems reuse the same primitives inside larger loops for indexing, retrieval, structure induction, verification, and delivery. For example, the diversity and coverage criteria used in representative image selection become useful for selecting evidence subsets before calling an MLLM; event segmentation and album organization become hypotheses about the latent event structure of G_C; and narrative planning or scene-graph-based storytelling anticipates claim planning and evidence-graph construction.

The continuity in Table 4 also clarifies the role of MLLMs. MLLMs change the interface and reasoning layer, but large visual contexts still require candidate selection, redundancy control, event grouping, metadata-aware indexing, temporal organization, and evidence attribution. The move to MLLM-based VMI is best understood as functional recontextualization: traditional primitives become reusable modules inside retrieval-augmented, evidence-constrained, and agentic reasoning workflows.

4.2 Limitations of Traditional Pipelines

Across the above task families, traditional approaches commonly introduce explicit structures before prediction. Image collections may be represented as hierarchies for scalable selection, clusters for event organization, rankings for representative image selection, sequences for temporal modeling, or graphs for object-relation reasoning. These structures make the problem tractable and interpretable: they reduce the search space, impose coverage or diversity constraints, and expose relations among images that are otherwise implicit. However, the same design also constrains systems to predefined objectives and output formats.

Traditional image sequence and photo collection methods provide important foundations, but they also have several limitations. First, most systems are task-specific: a method designed for photo selection cannot directly answer open-ended questions, and a narrative generation model is not necessarily able to ground evidence or compare arbitrary images. Second, many approaches depend heavily on handcrafted features, metadata, or constrained supervision, which may be unreliable in real collections. Third, their semantic abstraction is limited compared with recent MLLMs; complex entity relations, implicit events, user preferences, and evidential chains are difficult to model with fixed pipelines. Finally, traditional systems are largely passive: they do not actively inspect a collection, call external tools, retrieve supporting images, or maintain long-term memory.

These limitations motivate the move from collection-specific pipelines to MLLM-based multi-image understanding. Modern models provide a more general interface for reasoning over multiple images, following instructions, grounding answers, and integrating visual evidence with language. The move is still cumulative: the selection, grouping, ordering, and planning operations reviewed above remain necessary for indexing large collections, constructing candidate evidence, and controlling visual context. The next section reviews how recent MLLM research expands these foundations into more flexible multi-image reasoning systems.

5 Multi-Image Understanding in the MLLM Era

Multimodal large language models (MLLMs) move large visual contexts closer to general-purpose visual-language interaction. Earlier collection pipelines typically required separate models or heuristics for selection, summarization, storytelling, and retrieval. A single MLLM interface can in principle receive multiple images and natural-language instructions, then perform comparison, question answering, grounding, summarization, or dialogue. This unified interface is central to the move from traditional image-sequence analysis to modern visual macroscopic intelligence, but the harder question is whether it preserves cross-image correspondences, rejects distractors, reasons over temporal order, and selects evidence faithfully as the number and heterogeneity of images grow.

Table 5 summarizes representative directions in MLLM-based multi-image understanding. The table is organized by capability rather than by benchmark name, because the goal of this section is to characterize the methodological transition. A more detailed treatment of datasets and evaluation protocols is deferred to Section 8.

The detailed literature supporting each direction in Table 5 is provided in the supplementary material. The main text uses these directions to show a concrete boundary: MLLM research has

Table 5. Representative directions in MLLM-based multi-image understanding.

Direction	Representative objective	Core capability	Representative works
MLLM foundations	Connect visual encoders with language models and support visual instruction following.	General visual-language interaction; single-image reasoning; multimodal dialogue; grounding-aware alignment.	Flamingo, BLIP-2, LLaVA, Kosmos-2, Qwen2-VL, InternVL2.5 [39, 54–58]
Interleaved training	Train models to handle interleaved image-text contexts and multiple visual inputs.	Image identity preservation; co-reference; comparison; multi-image dialogue.	MANTIS, LLaVA-NeXT-Interleave, mPLUG-Owl3, MMDU, OBELICS [4, 5, 59–61]
QA and benchmarked reasoning	Answer questions whose evidence is distributed across multiple images.	Visual aggregation; multi-hop reasoning; robustness to distractors.	BLINK, MIRB, ReMI, MuirBench, MMIU, MMRB [6–8, 62–64]
Temporal / transformation reasoning	Test whether models understand order, change, and state transitions across images.	Event-order reasoning; before-after comparison; transformation inference.	Mementos, TempVS/Burn After Reading, VisualTrans [65–67]
Spatial and multi-view reasoning	Evaluate or improve geometric consistency across viewpoints, spatial layouts, and camera poses.	Cross-view correspondence; spatial relation reasoning; 3D scene awareness.	All-Angles Bench, VSI-Bench, SpatialVLM [9, 42, 68]
Grounding and correspondence	Locate evidence, compare images, or identify relations across visual inputs.	Pixel/region grounding; cross-image matching; visual cue linking; evidence localization.	PRIMA, MC-Bench, GeM-VG, MMVM [69–72]
Long-context modeling	Process long image sequences, mixed image-video contexts, or text-rich visual documents.	Long visual context; scalable visual tokens; sparse evidence search.	Gemini 1.5, MMNeedle, LongLLaVA, Visual Haystacks, visual-document RAG [12, 38, 40, 41, 73]
Reasoning post-training and domains	Improve or evaluate multi-image reasoning in specialized tasks and domains.	CoT supervision; RL-based visual comparison; medical grounding; math and quality comparison.	MAmmoTH-VL, Insight-V, MiCo, MedSG-Bench [74–77]

made comparison, co-reference, temporal reasoning, grounding, and long-context visual interaction measurable, while most evaluations still use small, query-driven settings with preselected evidence.

5.1 What Multi-Image MLLMs Solve and What They Leave Open

MLLM-based multi-image understanding solves a practical interface problem that earlier collection pipelines did not address well. A model can compare a small number of images, follow multi-image instructions, and produce explanations through one language interface, instead of requiring a separate task-specific model for each output format. Recent instruction-tuned models and benchmarks make capabilities such as co-reference, visual comparison, multi-hop question answering, and distractor robustness easier to measure. In this respect, the field has moved beyond treating multiple images as independent captions.

The limitation is visible in the evaluation setup. Most systems perform best when the relevant images are already provided, the question is already specified, and the answer can be compressed into a short text choice or span. They remain unreliable at deciding which images should be inspected, whether two visual instances are truly the same entity, which intermediate evidence is necessary, or when the available collection is insufficient. A model may answer a curated multi-image question correctly while still failing to build a reusable evidence state for an album, medical case, visual document corpus, or remote-sensing archive. Common failure modes include identity conflation, order sensitivity, shortcut use from captions or dataset priors, hallucinated cross-image links, and overconfident answers when distractor images contradict the apparent solution.

Benchmark design can hide these weaknesses. Small image counts, multiple-choice answers, synthetic layouts, and pre-filtered evidence may resemble VMI while bypassing retrieval, grouping, uncertainty, and provenance. Benchmarks that do not check which image or region supports an answer cannot distinguish genuine cross-image reasoning from lucky aggregation or language-only inference. For VMI, MLLMs are powerful reasoning modules that need surrounding mechanisms for evidence selection, correspondence, verification, and calibrated refusal.

6 Latent Structure Induction for Visual Macroscopic Intelligence

The previous section reviewed the common MLLM setting in which a model receives several images and responds to a user-specified question or instruction. VMI starts from less prepared inputs. A phone album may contain travel photos, screenshots, receipts, and repeated portraits without an explicit task; a medical study may combine current and historical scans; a remote-sensing archive may contain repeated observations of the same region; and a visual document repository may mix figures, tables, scanned pages, and screenshots. In each case, the system must determine meaningful units of analysis before producing an answer or report.

We treat latent structure induction over G_C as a core problem of VMI. The system must infer which images form candidate events, which visual instances refer to the same entity, which observations are temporally or spatially comparable, and which images can support or contradict a generated claim. These events, entities, and sub-contexts are uncertain hypotheses induced from metadata, visual similarity, OCR, retrieval results, and model predictions; they may need revision when new evidence or a different user goal is introduced. Clusters in this setting behave less like final partitions and more like editable hypotheses: an agentic VMI system may create, merge, split, relabel, or retire event and entity clusters as observations and tool outputs update the evidence graph. The process turns raw image collections into structured knowledge and user-facing outputs through representation building, grouping, cross-image comparison, evidence attribution, and synthesis; a more detailed pipeline view is provided in the supplementary material. Earlier image-mining research provides an important lineage [22], while VMI adds natural-language interaction, open-ended task discovery, multimodal reasoning, and faithful evidence construction over visual records. X-Cluster illustrates one recent step in this direction by discovering multiple natural-language grouping criteria and inducing semantic partitions over unstructured image collections [78]. In the VMI framing, such language-mediated semantic partitions are one form of latent structure; the broader evidence-graph view additionally requires entity and event links, provenance, and claim-level support.

6.1 From Query-Driven Understanding to Latent Structure Induction

The distinction between query-driven multi-image understanding and VMI is fundamental. In query-driven settings, the task is specified before inference: the model receives a question, a small number of images, and an expected answer format. VMI is more under-specified. The system may need to infer whether the useful output should be an event cluster, a timeline, a list of documents, a change report, or a set of images requiring human review.

The model's role changes accordingly. A query-driven model mainly answers; a macroscopic reasoning system first defines the subproblem. In the notation of Section 3.2, it must infer which part of G_C should be constructed and queried for the current goal. For example, in a personal album, the same collection could support a travel diary, a search index for receipts, or a summary of recurring family members. The challenge is to choose the right abstraction before generating language: an event graph for a diary, an evidence index for receipts, an entity graph for recurring people, or a memory graph for long-term personalization.

6.2 Reasoning Targets in Large Visual Contexts

The information that can be reasoned over and extracted from large visual contexts spans multiple semantic levels. At the *content level*, the system extracts people, objects, scenes, text, landmarks, products, and visual quality cues. These elements provide the atomic observations for later reasoning. At the *event level*, the system groups images into meaningful episodes such as trips, meetings, meals, celebrations, medical examinations, construction stages, or environmental observations. Event-level structure is often the first step toward making a large collection navigable.

At the *temporal and spatial level*, the system infers order, progression, repeated routines, geographic trajectories, or longitudinal changes. This is particularly important for before-after reasoning, medical follow-up, remote-sensing monitoring, and life-log analysis. At the *relational and evidential level*, the system discovers links among images: whether two images depict the same entity, whether one image provides a close-up of another, whether a screenshot contains textual evidence for a visual claim, or whether two images show a change in state. Such relations are crucial for faithful reasoning because claims must be tied to the images that support them.

Finally, at the *narrative and user level*, a system may derive stories, reports, personal memories, user preferences, or long-term behavioral patterns. This level connects VMI to human-facing applications. Producing a travel diary, organizing a medical case, or remembering recurring people requires the system to decide which observations should persist and which should remain local to the current task.

6.3 Core Capabilities for VMI

VMI requires capabilities that are broader than standard multi-image QA. *Representation building* maps images into visual embeddings, metadata records, OCR text, captions, or document-layout features. This requirement is related to image retrieval [13], but the retrieval target is often contextual: a user may search for "the receipt from the beach trip," which depends on event context as much as visual similarity. Large-scale vision-language pretraining and multimodal retrievers such as CLIP, BLIP, SigLIP, and ColPali provide foundations for this step [79–82]. Interactive query rewriting, composed image retrieval, visual RAG, and deep image search further show how retrieval can incorporate user intent and visual context at different evidence granularities [12, 14, 83, 84].

A second capability is *collection structuring*. Earlier work on album summarization, image importance, and curation already studied how to group photos into events, reduce near duplicates, and select representative images [1, 18, 85]. Recent language-guided collection organization extends this idea from fixed album operations to open-ended semantic discovery: for example, X-Cluster first proposes criteria such as activity, location, mood, or food-related attributes, and then groups the same image collection under each criterion [78]. Modern VMI systems can reuse such criteria proposal, grouping, and labeling operations as intermediate hypotheses inside broader analysis pipelines. In macroscopic settings, structuring defines the units over which later retrieval, summarization, comparison, and verification operate.

A third capability is *cross-image inference*: resolving entities, comparing states, recovering order, or identifying contradictions. This builds on multi-image and transformation reasoning benchmarks discussed in Section 5 [62, 63, 86], but must handle distractors and missing intermediate observations. Unlike structuring, the goal is not only to organize the collection, but to derive claims that depend on relationships across multiple images.

Finally, collection-level outputs must be grounded and usable. An inferred event, timeline, or report should point to supporting images or regions, especially in medical analysis, remote sensing, and visual-document settings. Textual summary generation from cohesive image sets provides an early example of this output layer [87], but broader macroscopic tasks may require a staged process: form groups, describe each group, compare groups, verify evidence, and only then produce an overall report.

6.4 Application Scenarios

Personal photo assistants are a natural application of VMI. A system can help users organize albums, identify trips and events, generate diaries, retrieve images by semantic memory, or summarize recurring people and places. This setting inherits concerns from lifelogging and personal-database research: the collection is longitudinal, weakly structured, and valuable precisely because it links images with time, activity, location, and memory cues [88, 89]. Recent work on deep image search makes this setting more explicit by treating retrieval over personal visual histories as an agentic exploration problem, where the system must use temporal, spatial, entity-level, and event-level context to locate images that may not be identifiable from isolated visual content alone [84]. Life-log, social multimedia, and wearable-camera scenarios similarly require extracting routines, activities, and long-term patterns from large volumes of weakly structured images, with large-scale resources such as YFCC100M providing natural user-photo hierarchies for studying visual histories [90]. Exploratory collection analysis also supports dataset auditing, social-media trend analysis, and generative-model bias diagnosis, where the goal is to discover which semantic dimensions and subgroups are present before deciding what downstream intervention or report is needed [78].

Domain-specific applications further highlight the need for VMI. In medical settings, multiple images may correspond to different views, modalities, or time points, and the system must compare them to identify changes or evidence; large radiology datasets such as MIMIC-CXR and CheXpert show how images, reports, uncertainty labels, and expert annotations become coupled evidence sources rather than independent images [91, 92]. In remote sensing, collections may contain repeated observations over geographic regions, where goals range from retrieving semantically matched scenes to detecting temporal changes or summarizing environmental events; this makes cross-modal alignment, geo-registration, sensor variation, and change localization part of the evidence schema [93–96]. In visual document repositories, useful information may be distributed across figures, tables, scanned pages, screenshots, slide decks, charts, and document order, requiring text-rich image understanding, layout-aware provenance, visual retrieval, and evidence synthesis [12, 97–100]. These document-like contexts are a distinctive VMI case because faithful outputs must preserve OCR uncertainty, page provenance, and links between generated claims and supporting visual evidence. A more detailed comparison of domain-specific macroscopic visual settings is provided in the supplementary material.

6.5 From Collection Organization to Macroscopic Judgment

VMI reframes visual understanding around the structure of the large visual context. Retrieval, clustering, metadata indexing, event grouping, and visual-document search can already make large collections more navigable; traditional album analysis also shows that representative selection and temporal organization are useful primitives. These methods reduce an unbounded image repository into candidate entities, events, relations, and evidence sets, giving VMI its operational substrate.

The harder step is judging whether the discovered structure supports the user's goal. A system may cluster an album, retrieve visually similar images, or summarize an event while still missing semantic continuity, selecting a receipt from the wrong trip, or producing a fluent report whose key claims lack image support. The hardest unsolved capabilities are latent task discovery, entity and event linking under ambiguity, evidence sufficiency estimation, contradiction handling, and deciding when a collection lacks enough information to support a conclusion.

Evaluation shortcuts can make the problem look easier than it is. A clean album with reliable timestamps, a retrieved top-k image set, or a fluency-scored summary does not test the same capabilities as a heterogeneous phone history with uncertain relevance and rare evidence. Common failure modes include salient-cluster bias, duplicate-heavy summaries, metadata shortcutting, missed minority events, and error propagation from early grouping decisions. Future systems should expose their inferred macroscopic structure and treat organization as a hypothesis to be verified, not as a fixed preprocessing step before generation.

7 Agentic Visual Reasoning

The preceding sections identify a systems problem for VMI: useful information may be split across images, metadata, OCR text, and relations that are not annotated in advance, and exhaustive inspection is often infeasible. We formulate *agentic visual reasoning* as sequential decision-making for constructing, querying, and verifying the latent evidence graph of a large visual context under a human- or application-defined objective. The term *agentic* is used here in the standard sense of a policy that observes a state, selects actions, receives observations from an environment or tool interface, updates its state, and decides when to stop. Tool use, memory, and verification are functional mechanisms for this policy, not definitions by themselves. For structure induction, an agent can propose candidate event, entity, and sub-context hypotheses, inspect uncertain links, invoke retrieval, OCR, grounding, or comparison tools, and update the evidence graph before producing the final artifact. Compared with static collection organization, the agentic setting makes reasoning iterative: the system chooses what to inspect next, which tools to invoke, which structural hypothesis to revise, and when the accumulated evidence is sufficient for grounded delivery.

7.1 Agentic Problem Setting

Given a large visual context $C = \{(I_i, m_i)\}_{i=1}^{N}$, where I_i denotes an image and m_i denotes optional metadata such as time, location, OCR text, captions, or user annotations, the task is to produce a structured output Y for a reasoning goal g. The goal g can be explicit, such as summarizing a personal album, constructing a travel timeline, detecting changes in remote-sensing observations, or generating a medical case report. It can also be exploratory, expressed by a seed instruction such as "find valuable information in this collection." In that case, the agent proposes candidate targets and turns them into operational subgoals. Agentic visual reasoning is goal-conditioned reasoning with a maintained evidence state, not unconstrained visual browsing.

A minimal task instance can be written as a partially observable interaction tuple

$$(C, g, \mathcal{T}, \mathcal{S}, \mathcal{A}, \mathcal{O}, B),$$

where the visual context and goal define the environment. Here, $\mathcal{T}$ provides available tools; $\mathcal{S}$ is the internal evidence state space; $\mathcal{A}$ contains retrieval, inspection, comparison, verification, memory, and delivery actions; $\mathcal{O}$ contains observations returned by images, metadata, users, or tools; and B denotes constraints such as computation budget, privacy policy, interaction cost, and attribution requirements. We reserve *agentic* for settings with four operational conditions: partial observability, budgeted action selection, provenance-carrying observations, and an explicit stopping condition. Tool use or retrieval alone is insufficient without a maintained evidence state. The desired output Y may include event clusters, entity lists, timelines, reports, evidence graphs, searchable indexes, or persistent memories. Traceability is required: generated claims should be linked to supporting images, regions, metadata, or tool observations.

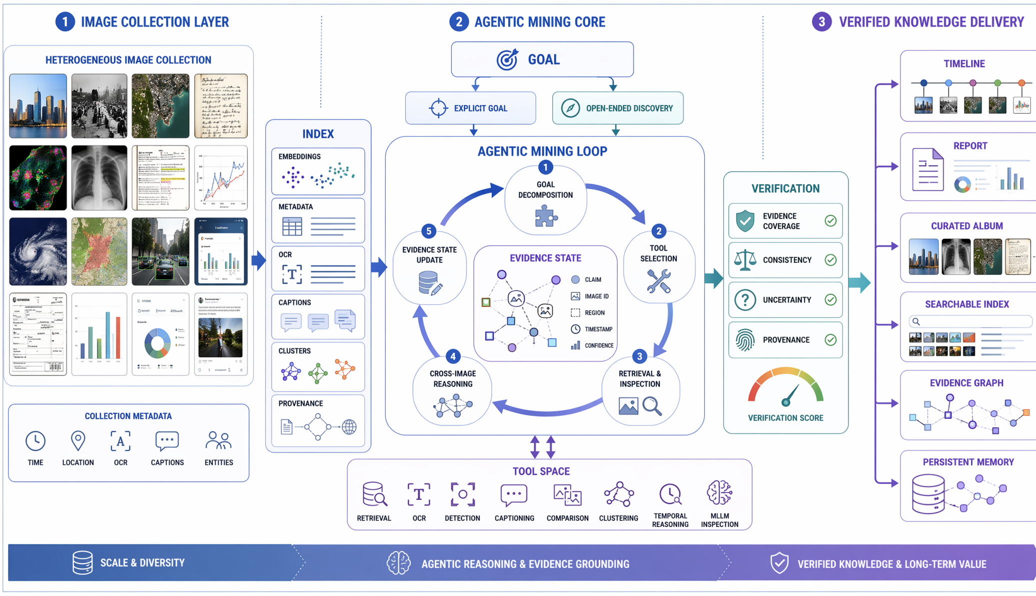

Fig. 4. Capability decomposition for agentic visual reasoning, covering context indexing, goal-conditioned tool use, iterative evidence-state reasoning, verification, and grounded knowledge delivery.

These requirements distinguish agentic visual reasoning from three related settings. It may locate the relevant subset before multi-image QA, continue beyond returning images in retrieval, and maintain a visual evidence state that general multimodal-agent work often leaves implicit: what has been found, why it matters, which observations support it, which hypotheses remain uncertain, and how confidence changes as new evidence is inspected.

7.2 Evidence-State Reasoning for Agentic Visual Reasoning

We summarize the methodology of agentic visual reasoning as an *evidence-state reasoning loop*. At step t, the agent maintains a state

$$s_t = (H_t, Z_t, Q_t, B_t),$$

where H_t denotes inspected evidence and tool observations, Z_t denotes the current hypothesis about the latent evidence graph, Q_t denotes unresolved questions or uncertain links, and B_t denotes constraints such as computation budget, privacy policy, interaction cost, and attribution requirements. The agent selects an action

$$a_t = \pi(s_t, g, \mathcal{T}),$$

where π is the agent policy and $\mathcal{T}$ is the available tool set. It then receives an observation o_t, updates the state, and stops when evidence is sufficient, the budget is exhausted, or the remaining uncertainty should be surfaced to the user. The loop connects to ReAct-style reasoning-action methods, but the state here persists beyond prompt-level action interleaving and records macroscopic visual evidence; related deliberation and self-correction paradigms are useful only when their intermediate states are grounded in retrievable visual evidence [10, 101, 102]. Existing retrieval, tool-use, visual-programming, and memory-augmented systems suggest an *Index–Goal–Tool–Reason–Verify–Deliver* capability decomposition. The decomposition is not tied to a specific model; it summarizes functional modules that repeatedly appear in systems for large-scale reasoning over visual contexts.

First, *indexing* converts raw images into searchable and traceable context representations. This module stores image identifiers, metadata, visual embeddings, OCR results, captions, document-layout features, entity candidates, quality signals, and coarse clusters. It provides the substrate for retrieval, deduplication, organization, and evidence citation.

Second, the system receives a *goal*. The goal may be an explicit user request, such as "summarize this album" or "find evidence of construction progress." It may also be an open-ended seed instruction, such as "find valuable information in this collection," under which the agent identifies candidate reasoning targets from the indexed context. The same capability pattern can be reused across goals because the agent maps either explicit goals or discovered targets into intermediate products: candidate events, entities, comparisons, missing evidence, output fields, or verification checks.

Third, the system exposes a *pluggable tool space*. A tool can be abstracted as

$$\tau_k : x_k \mapsto (o_k, \kappa_k, \gamma_k, \rho_k),$$

where x_k is the input, o_k is a structured observation, κ_k is the execution cost, γ_k is a confidence or uncertainty estimate, and ρ_k records provenance such as image IDs, regions, timestamps, or tool versions. This interface makes tool outputs usable as evidence rather than as ungrounded text. Minimal tools include retrieval, OCR, captioning, MLLM inspection, object or face detection, segmentation, referential grounding, clustering, pairwise comparison, temporal or geo-spatial reasoning, contradiction checking, and output generation; recent general-purpose segmentation and grounding tools make such modular evidence extraction more realistic, but they also need provenance and uncertainty records when used inside an agent [103–105].

Fourth, the agent performs *autonomous reasoning*. The abstract state $s_t = (H_t, Z_t, Q_t, B_t)$ is instantiated as concrete intermediate products: inspected image IDs, tentative claims, unresolved comparisons, tool outputs with confidence, and the remaining budget. The action a_t may query the index, select uncertain images, invoke OCR or detection, compare two sub-collections, or ask for human confirmation. After receiving an observation o_t, the agent updates the state:

$$s_{t+1} = U(s_t, a_t, o_t).$$

This loop replaces one-pass processing with selective, evidence-aware reasoning over the context.

Fifth, *verification* checks whether the inferred information is sufficiently supported. Verification may examine evidence coverage, cross-tool consistency, entity or event merging errors, temporal plausibility, contradictory images, and unsupported claims. Human feedback is not required for every reasoning task, but can serve as an optional verification mechanism when the system faces ambiguous identities, high-stakes domain conclusions, or sensitive inferences that should not be finalized solely by the agent.

Finally, *delivery* converts the verified state into a user-facing artifact or downstream action. The output may be a curated subset, timeline, album story, case report, inspection log, structured database, or reminder derived from the context. When appropriate, reusable information can be written back to memory for future tasks over the same collection or user history.

Table 6 summarizes this pattern as a capability decomposition rather than a prescribed architecture. Across the cited work, the common property is a controllable loop that keeps intermediate visual observations available for later checking and reuse.

7.3 Positioning Existing Work in the Capability Decomposition

Existing studies can be reinterpreted as partial evidence for this capability decomposition. ReAct establishes a prompt-level reasoning-action-observation loop, while Toolformer studies how language models can learn to call tools [10, 11]. These works provide the general agentic control pattern,

Table 6. A literature-derived capability decomposition for agentic visual reasoning.

Module	Function in agentic visual reasoning	Representative mechanisms	Main failure modes
Index	Build searchable, organized, and traceable representations of the visual context.	Embeddings; metadata/OCR indexing; caption caches; clustering; visual-document indexes; VisRAG-style retrieval [12, 100].	Missing metadata, biased embeddings, noisy OCR, weak provenance.
Goal	Specify an explicit objective or seed an open-ended search for valuable information, then map it into reasoning targets.	User instructions; open-ended seed prompts; candidate target proposal; task decomposition; evidence checklist; adaptive stopping criteria.	Ambiguous goals, low-value targets, over-decomposition, misaligned utility.
Tool	Provide specialized operations that can be invoked on images, regions, pairs, metadata, or retrieved subsets.	Retrieval; OCR; detection; face grouping; segmentation; geo-temporal analysis; visual programming [106–108].	Tool errors, incompatible interfaces, cost explosion, cascading failures.
Reason	Iteratively choose actions, inspect evidence, compare images, and update the reasoning state.	ReAct-style reasoning-action loops; controller agents; query rewriting; deep image search [10, 83, 84].	Query drift, missed evidence, inefficient exploration, spurious relations.
Verify	Check whether claims are sufficiently supported and internally consistent.	Image-level citation; region grounding; contradiction checking; uncertainty labels; human confirmation.	Hallucinated evidence, false confidence, unverifiable summaries.
Deliver	Produce a user-facing artifact or downstream action and optionally update reusable memory.	Curated image sets; summaries; timelines; alerts or reminders; case reports; evidence graphs; entity/event memory [109, 110].	Stale memory, privacy leakage, misleading abstraction, loss of uncertainty.

but their default problem setting does not maintain a persistent macroscopic visual state, evidence graph, provenance store, or memory-correction mechanism. Embodied and robotic systems such as SayCan, PaLM-E, and Voyager show how language-conditioned policies can connect perception, action, affordances, and open-ended exploration [111–113]. For VMI, their main lesson is not robotics itself, but the need to bind high-level plans to grounded observations and executable tools. Surveys of large multimodal agents further describe systems that integrate multimodal perception, planning, tool use, and interaction [16]. In contrast, the focus here is narrower but stricter: an agentic VMI system is evaluated by how its policy searches, updates, verifies, and delivers grounded knowledge from a large visual context, not merely by whether it can invoke tools during a dialogue.

Multimodal tool-use systems instantiate the *Tool* and *Reason* modules. Visual ChatGPT, HuggingGPT, Chameleon, VisProg, and ViperGPT show how language-model controllers can route subtasks to external visual experts or executable programs [106–108, 114, 115]. Most, however, are designed for single images, small image sets, or general multimodal interaction rather than persistent macroscopic reasoning.

Retrieval-centered methods mainly cover the *Index* and *Reason* capabilities. Interactive query rewriting, composed image retrieval, visual-document RAG, and VisRAG-style methods provide mechanisms for locating visual evidence in large repositories [12, 14, 83, 100]. Deep Image Search is especially close to agentic visual reasoning because it treats personal visual search as an iterative exploration process over contextual clues and image collections [84]. In the capability decomposition, it is a retrieval-centered case whose primary deliverable is a set of relevant images; VMI extends this direction toward verified outputs, evidence graphs, downstream actions, and reusable memory.

Autonomous visual information seeking also fits this view. AVIS demonstrates that an agent can decide which external resources to consult and how to refine visual information seeking based on intermediate observations [116]. In the capability decomposition, such work contributes to the policy for choosing reasoning actions. However, VMI additionally requires persistent indexing, evidence provenance, verification, and final synthesis over a large image repository.

Memory-augmented agents contribute to the *Deliver* and state-management components. Classical cognitive models distinguish short-term control processes from longer-term stores [109], while recent agent systems explore procedural memory and skill accumulation [110, 117]. For visual macroscopic intelligence, memory is useful only when it remains grounded: entity, event, and user-preference memories should preserve provenance, uncertainty, and correction mechanisms.

7.4 Benefits of the Agentic Reasoning Paradigm

The literature suggests that agentic reasoning is most useful when a large visual context is treated as a searchable evidence space that can be explored over time. This brings several practical advantages. First, it improves *scalability*: indexed representations and selective retrieval allow the system to reason over collections that exceed the visual context window of current MLLMs. Second, it enables *adaptive computation*: the agent can spend more effort on images that are uncertain, representative, contradictory, or central to a claim, while avoiding uniform processing of every item. Third, it supports *specialized perception and reasoning*: OCR, detection, face grouping, geo-spatial alignment, visual-document retrieval, and MLLM reasoning can be invoked only when relevant, instead of being compressed into a single monolithic model call.

The reliability benefit comes from keeping intermediate observations with provenance. Final claims can be traced back to specific images, regions, metadata fields, or tool outputs, which makes verification more concrete. The modular capability decomposition also allows different goals, tool sets, verification criteria, and output formats to be substituted without redesigning the whole system. Memory adds a cumulative dimension: repeated analysis of a personal album, domain archive, or visual document repository can gradually build grounded knowledge, instead of producing isolated summaries that are discarded after each interaction.

7.5 Agentic Reasoning: Gains, Failure Modes, and Accountability

Agentic visual reasoning addresses a concrete systems problem: no current MLLM can exhaustively inspect and reason over a large, heterogeneous visual context in one pass. Indexing, retrieval, tool use, state maintenance, and verification allocate computation selectively and keep intermediate observations available for later checking. The main gain over direct multi-image prompting is procedural: the system can search, compare, revise, and cite evidence before producing the final artifact.

The same procedure can fail in ways that answer-only benchmarks may not reveal. An agent may retrieve the wrong subset, follow an early mistaken hypothesis, call tools that confirm its assumptions, or stop before inspecting contradictory evidence. Common failure modes include query drift, visually salient but low-value exploration, cascading tool errors, circular self-verification, stale or overgeneralized memory, and privacy leakage from persistent indexes. These failures affect the reasoning trajectory and stored evidence state, not only the final response.

Current agent and visual-RAG evaluations may reward orchestration when the relevant evidence is easy to retrieve, tool outputs are treated as ground truth, or success is measured only by final answer accuracy. Harder evaluations should include missing evidence, conflicting observations, ambiguous identities, noisy OCR, irrelevant but visually attractive distractors, and tasks where abstaining or asking for clarification is correct. The unresolved capability is trustworthy policy: when to search, what to verify, how much evidence is enough, what to remember, and what should be forgotten.

Table 7. Capability coverage of proxy benchmark families for visual macroscopic intelligence. • denotes explicit evaluation, ◦ denotes partial or indirect coverage, and – denotes that the capability is absent. The final row summarizes the desired coverage of a full VMI benchmark rather than an existing resource.

Proxy benchmark family	Multi-image	Cross-image relation	Retrieval	Latent structure	Evidence grounding	Artifact output	Agentic process	Memory/privacy
Story / album / curation	◦	◦	–	◦	–	•	–	–
Open-ended semantic clustering	•	◦	–	•	–	◦	–	–
Multi-image QA / dialogue	•	•	–	–	◦	–	–	–
Temporal / transformation	•	•	–	◦	–	–	–	–
Multi-view / spatial	•	•	–	◦	◦	–	–	–
Grounding / correspondence	•	•	–	–	•	–	–	–
Long-context retrieval	•	◦	•	–	◦	–	–	–
Visual-document RAG	•	◦	•	–	•	◦	◦	–
Agentic visual search	•	◦	•	◦	◦	◦	•	◦
Full VMI benchmark	•	•	•	•	•	•	•	•

VMI systems should therefore be designed as accountable evidence systems. They should expose retrieved images, intermediate claims, provenance, uncertainty, and memory updates; distinguish observations from inferences; and allow users or domain experts to correct macroscopic structure. Without these mechanisms, agentic visual reasoning risks becoming a more complex way to hallucinate over larger visual repositories.

8 Datasets, Benchmarks, and Evaluation

Datasets and benchmarks determine how the community operationalizes visual macroscopic intelligence. They define the visual inputs, target outputs, and forms of reasoning that receive credit. Most existing resources evaluate partial capabilities needed for VMI, such as sequence-level narration, multi-image comparison, grounding, long-context retrieval, and visual-document evidence search. We treat them as *component-level proxy evaluations*: useful for diagnosing which parts of the VMI pipeline are currently measurable, yet insufficient for evaluating the full transformation from a weakly structured large visual context into a grounded, useful, and accountable knowledge artifact.

Table 7 provides a capability-level panorama of these proxy evaluations, while a representative resource landscape is provided in the supplementary material. The trajectory runs from task-specific evaluation on photo streams and albums to query-driven multi-image reasoning benchmarks for MLLMs, with newer resources beginning to expose the limits of answer-only evaluation through grounding, long-context retrieval, and agentic reasoning settings.

8.1 Traditional Sequence and Collection Datasets

Early resources evaluate image collections through concrete user-facing outputs. Visual storytelling datasets such as VIST ask a model to convert a photo stream into a coherent narrative rather than a set of independent captions [3]. RecipeQA extends this idea to procedural image-text sequences, where answering questions requires understanding multi-step cooking instructions across images [118]. Album summarization and photo selection tasks ask a model to retain representative, diverse, and important images from redundant collections [1, 2]. Multi-image summarization further studies textual summaries for cohesive image sets [87]. These tasks are valuable because they already treat the collection, rather than the individual image, as the unit of evaluation.

However, their evaluation protocols are usually output-specific. Storytelling and summarization systems are often judged by fluency, coherence, relevance, human preference, and automatic metrics borrowed from machine translation, text summarization, and image captioning, such as BLEU, ROUGE, and CIDEr [119–121]. Selection systems are judged by coverage, diversity, aesthetics, importance, or agreement with human-selected subsets. Such criteria capture important aspects of collection-level utility, but they provide limited supervision for evidence attribution. A story or summary can be fluent while introducing unsupported events, and a selected subset can be diverse while omitting images that are crucial for later reasoning.

Open-ended semantic clustering benchmarks make latent structure more explicit. COCO-4C and Food-4C annotate multiple grouping criteria and cluster labels for the same image collection, enabling evaluation of whether a system can discover natural-language organization principles and induce criterion-specific semantic partitions [78]. This is an important proxy for VMI because it tests structure discovery before downstream reasoning. However, it still covers only one output form: semantic partitions. It does not by itself evaluate claim-level provenance, evidence sufficiency, iterative verification, or whether the discovered structure supports a user-facing artifact such as a report, timeline, or memory.

8.2 MLLM-Based Multi-Image Benchmarks as Query-Driven Proxies

Recent MLLM benchmarks recast multi-image understanding as perception, question answering, and reasoning over several visual inputs. Earlier resources such as NLVR and Spot-the-Diff evaluated relational statements and image-pair differences [122, 123]. Newer benchmarks extend this setting to visual comparison, distributed evidence, distractors, unanswerable cases, multi-turn dialogue, and image-sequence reasoning, as in BLINK, MIRB, ReMI, MuirBench, MMIU, MMDU, MMMU, MM-Vet, and Mementos [6–8, 60, 62, 63, 65, 124, 125]. Spatial and embodied benchmarks such as ScanQA, SQA3D, and All-Angles Bench further test grounded 3D QA, spatial reasoning, and multi-view understanding [9, 126, 127].

These benchmarks reveal failure modes hidden in single-image VQA: incorrect entity binding, weak comparison, poor distractor rejection, hallucinated multi-hop evidence, and unreliable multi-view consistency. Yet they remain closer to query-driven reasoning than to VMI: the question is given, candidate evidence is usually included, and success is often answer accuracy. They therefore evaluate an important but incomplete slice of the corpus-reasoning pipeline.

8.3 Beyond Answer Accuracy: Evidence and Long-Context Evaluation

Answer-only benchmarks cannot determine whether a model used the correct visual evidence. Grounding-oriented resources address this limitation by requiring localization, correspondence, or region-level support. PRIMA studies multi-image pixel-grounded reasoning segmentation, while MC-Bench evaluates multi-context visual grounding across multiple images [69, 70]. Migician and GeM-VG extend the setting toward free-form and generalized multi-image grounding [71, 128]. Visual correspondence evaluations further show that strong MLLMs may fail to decide whether instances across images refer to the same entity [72].

Long-context and retrieval-oriented benchmarks evaluate another prerequisite for visual macroscopic intelligence: finding relevant evidence in a large visual context. SlideVQA, ChartQA, and multi-page DocVQA extend document-image evaluation toward slides, charts, and multi-page document images [99, 129, 130]. MMNeedle and Visual Haystacks test long multimodal context processing and visual needle retrieval [40, 41]. VisRAG and ViDoRAG evaluate visual-document retrieval-augmented generation, while DeepImageSearch introduces a benchmark for context-aware retrieval over personal visual histories [12, 84, 100]. These settings are closer to real large-scale visual contexts because they include irrelevant or weakly related visual evidence. Yet most of them

Table 8. Evaluation dimensions for visual macroscopic intelligence and agentic visual reasoning.

Dimension	Guiding question	Possible measurements	Typical failure modes
Coverage	Does the system represent the important parts of the collection?	Event/entity coverage, cluster recall, representative-image quality	Overfocusing on salient images; missing rare but important evidence.
Structure induction	Does the induced latent structure match the visual context?	Entity-linking accuracy, event grouping quality, graph-edge precision/recall, temporal violation count	Wrong merges or splits; missing latent events; noisy metadata shortcuts.
Cross-image reasoning	Can it infer relations distributed across images?	Co-reference accuracy, comparison accuracy, ordering accuracy, change-detection accuracy	Entity confusion, false temporal order, spurious before-after claims.
Evidence faithfulness	Are generated claims supported by correct visual evidence?	Citation precision/recall, region IoU or pointing accuracy, provenance correctness, contradiction rate	Hallucinated evidence, wrong image attribution, unsupported summaries.
Retrieval and organization	Can it find and structure useful sub-collections?	Retrieval recall, ranking quality, clustering purity, redundancy reduction	Missed evidence, query drift, duplicate-heavy outputs.
Synthesis utility	Is the final artifact useful for the user goal?	Human preference, task completion, report completeness, clarity and conciseness	Fluent but low-value summaries; loss of uncertainty or context.
Agentic process quality	Does the system reason over visual evidence efficiently and safely?	Tool-use cost, verification success, uncertainty calibration, memory consistency, privacy compliance	Cost explosion, cascading tool errors, stale memory, privacy leakage.

still emphasize locating a target or answering a specified query, rather than discovering latent structures, deciding what artifact should be produced, and generating a verified macroscopic output.

8.4 Evaluation Dimensions for Visual Macroscopic Intelligence

Visual macroscopic intelligence requires evaluation at the structure, output, and process levels. In the unified formulation, benchmarks should ask whether the induced structure Z is correct, whether each output claim has valid support E_c, whether the final artifact Y is useful for the goal, and whether the agentic policy π explores evidence efficiently and safely. The evidence constraint in Section 3.2 can be instantiated with computable proxies: if gold support sets E_c^* are available, claim support can be scored by citation precision and recall over image IDs or OCR spans; if regions are annotated, support can be scored by IoU or pointing accuracy; and if temporal or relational edges are annotated, structure can be scored by graph-edge precision/recall or temporal violation counts. Table 8 summarizes the resulting dimensions. The key point is that final-answer accuracy is insufficient: a system may answer correctly for the wrong reason, miss important sub-collections, or produce an appealing summary without traceable evidence.

Detailed metric operationalization, including structured traces, citation precision/recall, and an annotation-to-metric table, is provided in the supplementary material. The evaluation target should extend beyond questions and final answers. Useful benchmark records would include reference or partially annotated evidence graphs, macroscopic structure, evidence annotations, intermediate retrieval targets, contradiction cases, and human-usable outputs such as timelines, reports, summaries, or evidence graphs. For agentic systems, the reasoning trajectory itself should also be scored: which images were retrieved, which tools were called, which hypotheses in Z_t were revised, which claims were verified, and which uncertain inferences were withheld.

8.5 Gaps in Current Evaluation

Current evaluation remains incomplete for the full scope of visual macroscopic intelligence. Existing benchmarks are mostly proxy tests for individual capabilities: they are often small-scale, query-driven, weakly annotated for latent structure, and focused on answer correctness instead of evidence support or user utility. As a result, they rarely test whether a system can transform a weakly structured large visual context into a grounded and accountable artifact; privacy, consent, and memory correction also remain largely absent from benchmark design.

Hierarchical benchmarks are needed to require both final artifacts and evidence traces. Such benchmarks should combine large visual contexts, user goals, partial latent-structure annotations, claim-level evidence, distractors, uncertainty cases, and human-usable outputs such as timelines, reports, summaries, or evidence graphs. For agentic systems, they should also evaluate retrieval decisions, tool calls, hypothesis revisions, verification steps, and memory updates, making it possible to separate retrieval, grouping, perception, grounding, safety, and presentation failures.

9 Challenges and Future Directions

Visual macroscopic intelligence remains at an early stage. Recent MLLMs can process multiple images, but real-world visual contexts are heterogeneous, long-tailed, and weakly structured, requiring more than question answering over a few provided images. Recent work also frames images as external thinking media for search, comparison, and intermediate reasoning [131]. Under the latent evidence-graph view, progress depends on scalable evidence search over C, reliable structure induction Z, claim-level grounding in support E_c, and controllable agentic policies π for inspection, verification, memory, and forgetting. The following subsections organize these requirements around scalability, structure discovery, faithfulness, agent control, memory, and evaluation.

9.1 Scalable Long-Context Collection Modeling

A central challenge is scale. Most multi-image benchmarks contain only a few images, whereas real collections may contain hundreds or thousands of photos, screenshots, document pages, or domain observations. Directly feeding all images into an MLLM is expensive and often ineffective: visual token budgets are limited, irrelevant images distract the model, and useful evidence may be sparse. An album assistant may need to find one receipt or rare event among many daily photos, while a medical assistant may need to compare a current scan with selected historical examinations. Long-context models and benchmarks such as Gemini 1.5, LongLLaVA, and MMNeedle show progress toward longer visual contexts [38, 40, 73]. VMI still requires decisions about which images to inspect, summarize, retrieve again, or ignore.

Hierarchical collection representation is a concrete design direction. Future systems may build multi-level visual, textual, metadata, and entity/event indexes instead of encoding all images at one granularity. Coarse representations can support retrieval and clustering, while fine-grained inspection is invoked only for selected subsets. Walking-assistant systems similarly show that reducing redundant observations and outputs is important for timely VLM-based assistance in continuous streams [132]. Such architectures combine retrieval scalability with MLLM reasoning.

9.2 Cross-Image Correspondence and Structure Discovery

Another fundamental challenge is discovering latent structures across images. Collections often contain repeated people, objects, locations, documents, and events whose links are unannotated and visually ambiguous: the same entity may change across viewpoint, lighting, resolution, time, or occlusion, while similar images may denote different products, landmarks, or events. Recent evaluations show that strong MLLMs still struggle with visual correspondence and cross-image matching [72], and All-Angles Bench reports failures in occluded cross-view correspondence and coarse camera-pose reasoning [9]. These errors propagate to album organization, temporal reasoning, evidence aggregation, and memory construction.

Future work should build explicit mechanisms for entity linking, event grouping, temporal ordering, and relation-graph construction. For overlapping views, these mechanisms should incorporate geometric constraints such as pose uncertainty, epipolar consistency, visibility, occlusion, scale changes, and map persistence. Rather than treating images as independent context items, models should construct collection-level structures—entity graphs, event timelines, evidence chains, contradiction links, and spatial correspondence graphs—as interpretable substrates for retrieval, summarization, and verification.

9.3 Faithful and Evidence-Grounded Reasoning

Faithfulness is especially difficult in multi-image settings. A model may answer correctly for the wrong reason, merge evidence from different images, attribute a fact to the wrong source, or hallucinate relations that are not visually supported. In an album-summary setting, for instance, a model may generate the claim that "this is the second day of the trip" even though the supporting images actually come from different locations or different dates; in a visual-document collection, it may cite a chart image to support a conclusion that is only stated in a separate table screenshot. These risks are amplified in collection-level generation, where the output may be a story, timeline, report, or memory rather than a short answer. Grounding-oriented benchmarks such as PRIMA, MC-Bench, and GeM-VG begin to address this issue by requiring evidence localization or multi-image grounding [69–71], but fine-grained evidence tracking remains far from solved.

Future systems should treat evidence as a first-class object. Every generated claim should ideally be linked to supporting image IDs, regions, metadata fields, OCR snippets, or tool observations. Verification modules should check whether the evidence actually supports the claim, whether alternative evidence contradicts it, and whether the system should express uncertainty. This is particularly important for medical, remote-sensing, legal, and personal-memory applications, where unsupported conclusions may cause real harm.

9.4 Open-Ended Reasoning and Agentic Control

Most current benchmarks are query-driven: the user asks a question, and the system answers it using provided images. Real large visual contexts often require a more open-ended process. A user may simply ask the system to organize an album, find important events, summarize a trip, identify changes over time, or extract useful information from a repository. In such cases, the system must decide what is worth discovering before it can answer. For example, given a mixed phone album, an agent may need to decide whether the most useful output is a travel timeline, a set of document screenshots, a list of recurring people, or a warning about duplicate and low-quality photos. The problem becomes active, goal-conditioned visual reasoning rather than passive multi-image understanding.

Agentic systems offer a natural direction because they can turn open-ended requests into stepwise exploration policies. Work on ReAct, visual RAG, and context-aware image search suggests that tool-augmented agents can navigate visual evidence spaces more flexibly than one-shot models [10, 12, 84]. However, agentic visual reasoning also introduces new challenges: inefficient exploration, query drift, tool failures, cascading errors, and unclear stopping criteria. A visually salient but low-value cluster, such as many food photos, may attract repeated inspection, while a small set of receipt or medical images that matters more to the user may be missed. Future research should study compact reasoning states, budget-aware action selection, intermediate claim checking, and user-facing process transparency.

9.5 Memory, Personalization, and Privacy

Image collections are often personal. They may contain faces, locations, documents, health information, habits, routines, and sensitive events. At the same time, many valuable applications require memory: a personal assistant should remember recurring people, places, preferences, and past interactions; a domain assistant should accumulate knowledge about cases, sites, or longitudinal observations. This creates concrete privacy risks: a system may write a face identity, hospital receipt, home address, school location, or family routine into long-term memory, and later reuse it in an unrelated task or expose it to the wrong user. Memory-augmented agents and continual multimodal systems point toward this direction [110, 117], but large visual context memory must remain grounded, editable, and privacy-preserving.

Future systems need mechanisms for consent-aware indexing, access control, selective forgetting, uncertainty-aware memory, and user correction. A memory should not simply store generated text; it should preserve provenance, confidence, timestamps, and links to supporting evidence. Users should be able to inspect why a memory was created, correct mistaken entity associations, and delete sensitive information. Without such controls, macroscopic reasoning may become intrusive or unreliable.

9.6 Toward Unified Benchmarks and Real-World Deployment

Finally, the field needs benchmarks that reflect the full VMI pipeline rather than isolated components such as storytelling, album summarization, multi-image QA, grounding, long-context retrieval, visual RAG, or context-aware image search. Future benchmarks should require systems to convert large visual contexts into grounded artifacts—for example, timelines, progress reports, or follow-up summaries with cited visual evidence—and should evaluate both the final output and intermediate retrieval, context coverage, evidence faithfulness, process efficiency, and user utility.

Real-world deployment further requires domain-specific adaptation. Personal albums, medical image sets, remote-sensing archives, visual documents, and e-commerce galleries differ in structure, metadata, risk, and evidence requirements. Modular systems should therefore adapt their indexing, tools, verification criteria, evidence schemas, and output formats to each domain while preserving a common macroscopic visual framework. In this sense, VMI should evolve from isolated multi-image reasoning tasks into a general paradigm for organizing, reasoning over, and delivering grounded knowledge from large visual contexts.

10 Conclusion

This survey has reviewed visual macroscopic intelligence as a task and research paradigm for interpreting large visual contexts, where useful information is distributed across images through recurring entities, events, temporal-spatial relations, evidential links, and user context. We argued that tasks such as selection, summarization, storytelling, question answering, retrieval, report generation, and memory construction can be unified as goal-conditioned outputs over a shared latent macroscopic structure.

To make this view operational, we used a latent evidence-graph abstraction that links images, entities, events, claims, memories, and provenance under evidence constraints. We connected traditional photo-collection analysis, MLLM-based multi-image understanding, latent structure induction, datasets and benchmarks, and agentic visual reasoning under this abstraction. The literature on agents, retrieval, tools, grounding, and memory suggests a path beyond one-shot multi-image QA toward active evidence search, hypothesis testing, verification, and grounded delivery. Future progress will depend on scalable collection modeling, robust cross-image correspondence, faithful evidence grounding, controllable agentic workflows, privacy-preserving memory, and evaluation protocols that reflect real-world macroscopic visual use.

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A Methodology, Scope, and Survey Positioning

A.1 Survey Methodology and Corpus Construction

This survey is designed as a structured scoping survey rather than an exhaustive systematic review. Its goal is to map the emerging research space around visual macroscopic intelligence, identify recurring task formulations and system capabilities, and synthesize representative methods across several partially overlapping areas, including image collection understanding, multi-image reasoning, visual grounding, long-context multimodal modeling, retrieval-augmented visual reasoning, and multimodal agents. This design follows the role of scoping and mapping reviews in clarifying concepts, boundaries, and evidence landscapes for emerging fields, rather than estimating a single effect size or enforcing a narrow intervention protocol [19–21, 133, 134]. We therefore discuss adjacent fields selectively when they contribute to macroscopic visual organization, reasoning, evidence attribution, or grounded knowledge construction, rather than attempting to cover all work in image retrieval, video understanding, or agents.

We constructed the paper set through a multi-stage literature search covering work published or released up to May 2026. Candidate papers were collected from arXiv, Google Scholar, ACM Digital Library, IEEE Xplore, CVF Open Access, ACL Anthology, and DBLP. The search used combinations of keywords including *visual macroscopic intelligence*, *image collection understanding*, *image mining*, *multimedia data mining*, *content-based image retrieval*, *multi-image understanding*, *image sequence understanding*, *photo album summarization*, *visual storytelling*, *multi-image question answering*, *multi-image grounding*, *multi-view geometry*, *structure from motion*, *visual SLAM*, *long-context multimodal models*, *visual retrieval-augmented generation*, *agentic visual reasoning*, and *multimodal agents*. Rather than treating keyword retrieval as a complete census, we used it as a seed set for iterative concept expansion: backward reference tracing exposed older collection-organization and retrieval foundations, while forward citation tracing from datasets and benchmarks identified newer multi-image, grounding, and visual-RAG work that often uses different terminology. Because this is a scoping survey over an emerging and terminology-fragmented area, raw platform hit counts were not used as evidence of coverage: Google Scholar and arXiv counts are unstable over time, and relevant papers are often indexed under neighboring labels such as visual storytelling, visual RAG, visual correspondence, document-image reasoning, or multimodal agents. Instead, we maintained a deduplicated candidate pool using title, DOI, arXiv identifier, venue, and publication year, and screened each candidate for its contribution to macroscopic visual organization, cross-image reasoning, evidence attribution, or agentic visual evidence handling.

Table 9 makes the search and screening protocol explicit. The protocol is intended to support reproducibility in future updates: readers can rerun the same query families on the same sources, merge backward and forward citation results, remove duplicates by bibliographic identifiers, and then apply the boundary criteria in Table 10. The resulting set should be interpreted as an analytically representative scoping corpus rather than a statistically exhaustive bibliometric census.

A.2 Limitations of This Survey

This survey has several limitations. First, it is a structured scoping survey rather than an exhaustive systematic review. Its goal is to map concepts, task formulations, capabilities, and evaluation gaps around VMI, not to provide a complete bibliometric census of all work in image retrieval, video understanding, MLLMs, visual RAG, and multimodal agents. Second, the literature boundary is unstable because many relevant 2025–2026 works are preprints, technical reports, or rapidly updated benchmark releases; recent models, datasets, and leaderboard results may therefore be missing or may change after publication. Third, benchmark statistics are not fully comparable across papers: different resources report images, frames, pages, slides, questions, dialogues, visual histories, or retrieval units under different conventions. We use these numbers mainly to characterize input scale and evaluation scope rather than to rank benchmark difficulty. Finally, because VMI intentionally connects several adjacent areas, boundary cases exist between VMI and neighboring topics such as image retrieval, video understanding, visual RAG, and multimodal agents. We address this by making the boundary criteria explicit through inclusion criteria, comparison tables, and supplementary coding protocols, rather than treating VMI as a closed category.

A.3 Search Protocol and Boundary Criteria

This subsection expands the compact scoping flow in the main paper into the concrete protocol used for search, screening, and coding. Table 9 separates the sources, screening actions, and recorded outputs for each stage, while the text below states the query families and boundary rules used to keep the corpus focused on VMI-relevant work.

The seed query families were organized into six groups. The first group targeted traditional collection analysis and retrieval, using terms such as *image mining*, *multimedia data mining*, *content-based image retrieval*, *photo album summarization*, *image collection organization*, and *visual storytelling*. The second group targeted MLLM-based multi-image understanding with terms such as *multi-image understanding*, *multi-image question answering*, *interleaved image-text instruction tuning*, *multi-image dialogue*, and *image sequence reasoning*. The third group targeted relation and evidence modeling, including *multi-image grounding*, *visual correspondence*, *cross-image matching*, *temporal visual reasoning*, and *visual transformation reasoning*. The fourth group covered long-context and retrieval-augmented settings, including *long-context multimodal models*, *visual haystack*, *visual retrieval-augmented generation*, *visual-document RAG*, and *context-aware image search*. The fifth group covered spatial and multi-view foundations, including *multi-view geometry*, *structure from motion*, *visual SLAM*, *multi-view reasoning*, and *spatial visual reasoning*. The sixth group covered agentic mechanisms, including *agentic visual reasoning*, *visual information seeking*, *multimodal agents*, *visual tool use*, and *memory-augmented multimodal agents*.

Papers were included when they satisfied at least one of the following criteria: (1) they treat a sequence, set, or repository of images as the main input; (2) they produce a macroscopic output such as a representative subset, event grouping, story, report, grounded answer, or benchmark label; (3) they study MLLMs under multi-image, long-context, grounding, or evidence-search settings; or (4) they introduce retrieval, tool-use, state, memory, or agent mechanisms that can support collection-level visual analysis and reasoning. We excluded pure single-image recognition, captioning, or VQA except when they serve as foundations for multi-image reasoning; dense video understanding unless it informs sparse image-sequence or collection-level reasoning; generic image retrieval unless it contributes to evidence search or collection organization; and general-purpose agents without explicit visual evidence handling. The resulting paper set is therefore intended to be analytically representative: it supports a taxonomy of VMI capabilities and failure modes while acknowledging that neighboring areas have their own specialized surveys.

The final paper set was curated to cover historically influential tasks, widely used datasets and benchmarks, technically distinct modeling paradigms, and recent boundary-expanding systems. We then derived the taxonomy iteratively from the screened papers by grouping them according to input structure, reasoning capability, system role, and output form. For each included work, we coded six attributes: publication year, dominant role in the survey, input structure, primary capability, evaluation setting, and VMI relevance. These attributes support the quantitative development landscape in Figure 5 and the capability-oriented tables in the main paper and appendix. Table 10 summarizes the resulting boundary and explains why adjacent areas such as visual storytelling, image retrieval, video understanding, and multimodal agents are discussed selectively rather than exhaustively.

Table 9. Search and screening protocol used to construct the scoping corpus. The protocol emphasizes reproducible query families, citation tracing, deduplication, and capability-oriented coding rather than unstable raw hit counts.

Stage	Sources or inputs	Screening action	Recorded output
Seed keyword search	arXiv, Google Scholar, ACM Digital Library, IEEE Xplore, CVF Open Access, ACL Anthology, and DBLP.	Search query families around image collections, multi-image understanding, grounding, long-context multimodal models, visual RAG, multi-view reasoning, and multimodal agents.	Candidate records with title, authors, venue or preprint source, year, DOI or arXiv identifier, and matched query family.
Backward tracing	References of seed papers, datasets, benchmarks, and related surveys.	Add historically influential work that established collection organization, image retrieval, visual storytelling, multi-view correspondence, document-image reasoning, or agentic tool use.	Older foundational records that may not use current VMI or MLLM terminology.
Forward tracing	Citation links from major datasets, benchmarks, and representative systems.	Add recent systems, benchmarks, and follow-up studies that extend multi-image reasoning, grounding, visual RAG, or agentic visual evidence search.	Boundary-expanding records, especially 2024–2026 work released under rapidly changing terminology.
Deduplication	Combined candidate pool.	Merge records by title normalization, DOI, arXiv identifier, and venue version; prefer peer-reviewed versions when a preprint and formal version both exist.	Deduplicated candidate pool for screening and coding.
Eligibility screening	Deduplicated records and full texts when needed.	Apply the inclusion and exclusion criteria below, retaining work that contributes to macroscopic visual organization, cross-image reasoning, evidence construction, evaluation, or agentic visual reasoning.	Included papers plus exclusion notes for single-image-only, dense-video-only, generic-retrieval-only, or non-visual-agent-only papers.
Capability coding	Included papers.	Code each paper by dominant role, input structure, reasoning capability, system role, output form, evaluation resource, and VMI limitation.	Taxonomy, quantitative trend categories, boundary table, benchmark landscape, and capability-level synthesis.

A.4 Positioning Against Related Survey Lines

This subsection clarifies how the survey differs from neighboring survey lines. The comparison is not intended to replace specialized surveys on image retrieval, composed retrieval, multimodal RAG, MLLMs, or multimodal agents; instead, it identifies the part of each area that contributes to VMI and the part that remains outside their usual scope. Table 11 summarizes these distinctions.

B Conceptual Framework and Development Landscape

B.1 Orthogonal Taxonomy for Visual Macroscopic Intelligence

A recurring risk in survey papers is that the taxonomy becomes a catalog of model names, dataset names, or historical subfields. To avoid this, we use an orthogonal taxonomy that codes each work

Table 10. Boundary criteria for including adjacent areas in this survey. Adjacent areas are included only when they contribute to macroscopic visual organization, latent structure induction, evidence attribution, or grounded knowledge construction.

Adjacent area	Included in this survey when	Not treated as main focus	Role in this survey
Single-image VLMs	They provide foundations for perception, captioning, grounding, or visual instruction following.	Exhaustive review of single-image recognition, captioning, or VQA.	Background for multi-image and macroscopic MLLMs.
Video understanding	Sparse image-sequence, temporal reasoning, or long-context settings inform macroscopic reasoning.	Dense action recognition or frame-level video modeling as a standalone field.	Contrastive paradigm for ordered visual context.
Visual storytelling and album analysis	The task requires selection, ordering, summarization, or narrative abstraction over photo streams.	Stylistic language generation detached from collection structure.	Traditional precursor of visual macroscopic intelligence.
Image and multimedia mining	They provide early mechanisms for extracting patterns, semantics, clusters, and summaries from image repositories.	General multimedia mining detached from visual evidence attribution or macroscopic artifacts.	Historical technical lineage for indexing, clustering, association discovery, and scalable collection analysis.
Multi-view geometry and robotic perception	Cross-view matching, 3D reconstruction, or SLAM informs correspondence, pose, and spatial evidence across images.	Low-level geometry as a standalone reconstruction problem without semantic or user-facing synthesis.	Geometric foundation for cross-image correspondence, spatial consistency, and view-grounded evidence.
Image retrieval and visual RAG	Retrieval supports indexing, evidence search, visual-document reasoning, or context-aware image search.	Generic instance-level retrieval without evidence construction or macroscopic synthesis.	Infrastructure for evidence selection and grounded generation.
Multimodal agents	Agents actively retrieve, inspect, or verify visual evidence over collections.	General language-agent frameworks without visual evidence state or macroscopic provenance.	Methodological instantiation within VMI.
Domain applications	Medical, remote-sensing, document, or personal-photo settings require multi-image comparison or macroscopic synthesis.	Domain-specific low-level processing without cross-image semantics.	Application drivers and evaluation scenarios.

by the problem dimensions it addresses rather than by the label under which it was originally published. This taxonomy complements the main-paper organization: the main text follows a readable progression from traditional collection analysis to MLLM-based multi-image understanding, latent structure induction, agentic visual reasoning, and evaluation, while the dimensions below explain the cross-cutting axes used to compare works across these sections.

Table 12 should be read as a coding scheme rather than as a sequence of sections. For example, MLLM multi-image QA benchmarks are usually small- to medium-scale, highly task-specified, answer-oriented, and evaluated by final accuracy, even when they contain multiple images. Traditional album summarization is often collection-level and artifact-oriented, but usually has fixed output forms and limited claim-level provenance. Visual-document RAG and visual-history search improve retrieval and evidence access, but may still lack latent structure induction or artifact-level synthesis. Agentic visual reasoning adds an explicit process axis by deciding what to retrieve,

Table 11. Comparison with related surveys and survey lines. The table highlights why this survey treats large weakly structured visual contexts as a distinct object of synthesis rather than as only retrieval input, MLLM context, or agent observations.

Related survey line	Primary coverage	Remaining gap for VMI	This survey
Image retrieval [13]	Representation, indexing, similarity search, relevance feedback, and ranking.	Treats collections mainly as searchable resources rather than latent structures with claims and provenance.	Recasts retrieval as evidence access, sub-context selection, deduplication, and scalable exploration inside macroscopic reasoning.
Composed image retrieval [14]	Image-text query composition, retrieval models, datasets, and ranking evaluation.	Focuses on query-conditioned ranking, not open-ended structure discovery or artifact generation from collections.	Connects composed retrieval to goal-conditioned evidence selection and cross-image relation construction.
Multimodal / vision RAG [135, 136]	Retrieval-augmented pipelines for multimodal QA, visual understanding, and generation.	Emphasizes retrieved evidence and generation, but less the induction of entities, events, contradictions, and reusable visual evidence states.	Places visual RAG within a broader evidence-graph process for latent structure induction, verification, and grounded artifacts.
General MLLM surveys [137]	Architectures, training, modalities, generation abilities, and general evaluation of MLLMs.	Does not center weakly structured image collections, cross-image provenance, or collection-level outputs.	Uses MLLMs as reasoning modules within VMI systems that also require indexing, structure discovery, evidence grounding, and evaluation.
Multimodal-agent surveys [16]	Planning, tool use, memory, collaboration, and general multimodal agent applications.	Agent state is not specialized for large visual contexts, visual correspondence, provenance, or claim-level support.	Defines agentic visual reasoning as evidence-state construction over large visual contexts with verification, memory control, and artifact delivery.

inspect, verify, and remember, but it still must be evaluated for evidence faithfulness and privacy rather than for tool use alone.

This orthogonal view is the basis for the capability-oriented tables in the main paper. The traditional-collection section mainly covers collection organization and artifact generation under relatively fixed objectives; the MLLM section covers query-driven multi-image reasoning and its limits; the latent-structure section focuses on structural uncertainty and evidence induction; the agentic-reasoning section focuses on the system-process axis; and the evaluation section turns the same axes into benchmark and metric requirements.

Using the taxonomy as a diagnostic key. For a new paper, the taxonomy can be applied through three compact checks. First, the *context key* asks whether the relevant images and question are already supplied; if not, the work should be coded as retrieval-, indexing-, or structure-induction oriented, and should report evidence such as context coverage or retrieval recall. Second, the *structure and evidence key* asks whether entities, events, temporal order, and support links are given or induced; induced structure requires cluster, link, timeline, or provenance evidence rather than answer accuracy alone. Third, the *process and artifact key* asks whether the system maintains a

Table 12. Orthogonal taxonomy for organizing visual macroscopic intelligence literature. Each axis can be coded independently, allowing works to be compared by problem properties rather than by model family, benchmark name, or application domain.

Axis	Typical values	Guiding question	Why it matters for VMI
Visual-context scale	Image pair; small image set; ordered sequence; long visual context; repository or personal history.	How much visual evidence must be considered, and can it fit in a single model context?	Determines whether direct prompting is sufficient or whether retrieval, compression, indexing, and hierarchical representation are required.
Structural uncertainty	Predefined input structure; weak metadata; noisy order; latent entities or events; heterogeneous mixed collections.	Are the relevant units and relations already given, or must they be discovered?	Separates ordinary multi-image answering from event grouping, entity linking, temporal-spatial organization, and evidence-graph induction.
Task specification	Fixed question; instruction-following; goal-conditioned artifact; exploratory discovery; continual memory update.	Is the target known before inference, or must the system formulate useful subgoals?	Distinguishes answer generation from open-ended macroscopic reasoning, where the system must first decide what structure or artifact is useful.
Evidence requirement	Answer-only; image-level citation; region or OCR grounding; cross-image correspondence; claim-level provenance graph.	What support must be attached to the output?	Controls faithfulness: VMI outputs such as timelines, reports, and memories require traceable support rather than fluent unsupported summaries.
Output artifact	Answer; representative subset; cluster or index; ordered story; timeline; report; evidence graph; persistent memory.	What user-facing object is produced from the collection?	Prevents evaluation from collapsing to accuracy alone and links methods to collection-level utility.
System process	One-shot MLLM inference; retrieval-augmented generation; tool-augmented inspection; iterative agentic reasoning; human-in-the-loop verification.	How is evidence selected, inspected, revised, and verified?	Captures the difference between processing a provided set and actively constructing an evidence state over a large visual context.
Evaluation target	Final-answer correctness; structure quality; retrieval recall; evidence faithfulness; artifact utility; process efficiency; privacy or memory safety.	Which part of the VMI pipeline receives credit?	Makes benchmark limitations explicit and reveals when a dataset measures only a proxy capability rather than full macroscopic intelligence.

budgeted, provenance-aware evidence state with an explicit stopping condition, and what artifact it produces. These keys help diagnose common gaps: accuracy-only work often lacks faithfulness evidence, retrieval-only work lacks artifact utility, clustering-only work lacks claim provenance, and tool-use work lacks process or stopping-condition evaluation.

B.2 Quantitative Development Landscape

To complement the qualitative synthesis, we provide a quantitative landscape of the cited literature. Each cited work is coded by publication year and by its dominant role in this survey: traditional and adjacent macroscopic visual foundations, MLLM-based multi-image understanding, or VMI-aligned macroscopic reasoning. The first category covers image and multimedia mining, retrieval, grounding datasets, multi-view geometry, photo collection analysis, visual storytelling, document/chart/medical/remote-sensing foundations, review methodology, and general tool-use or embodied-agent foundations that do not by themselves construct macroscopic visual evidence, with representative roots in image mining, content-based image retrieval, and visual storytelling [3, 22, 43]. The second category covers VLM/MLLM models, multi-image instruction tuning, multi-image benchmarks, spatial reasoning, hallucination, and reasoning post-training, as represented by recent multi-image models and benchmarks [4, 5, 7, 8]. The third category is reserved for work that more directly supports macroscopic evidence construction, including long-context visual evidence search, visual-document RAG, multi-image grounding and correspondence, domain-specific sequence and collection benchmarks, visual-history search, memory, and agentic evidence exploration [12, 71, 72, 84, 100]. This stricter coding avoids inflating VMI-aligned counts with generic 2023 agent papers. The plot is not intended as an exhaustive bibliometric study; rather, it summarizes how the cited evidence base shifts from earlier macroscopic visual foundations toward recent MLLM and VMI-aligned work. A single pre-1999 bin is used for one classic memory reference, and the 2026.5 bin covers work released up to May 2026.

Figure 5 shows three broad trends. First, earlier references are dominated by macroscopic visual and adjacent foundations, including image and multimedia mining, content-based retrieval, scalable visual search, visual grounding, multi-view correspondence, photo collection organization, document/chart reasoning, domain datasets, summarization, storytelling, and general agent/tool-use foundations. Second, the cited literature becomes denser from the mid-2010s to the early 2020s as visual grounding resources, document-image reasoning, multimodal representation learning, and domain-specific visual datasets mature. Third, after 2023 the field separates into two fast-growing streams: MLLM-based multi-image understanding expands rapidly through models and benchmarks, while VMI-aligned work grows more specifically around long-context visual evidence search, visual-document RAG, multi-image grounding and correspondence, domain-specific collection benchmarks, memory, context-aware image search, and agentic evidence-state reasoning.

B.3 Supplementary Pipeline View of Latent Structure Induction

Figure 6 expands the process view behind latent structure induction for VMI. The purpose of this pipeline is not to introduce another independent framework, but to make explicit the intermediate operations that connect the conceptual definition of VMI with concrete system design. Starting from unordered or weakly structured image collections, the system first builds traceable representations through metadata, OCR, visual features, embeddings, and indexes. It then organizes the collection into candidate clusters, event groups, and representative subsets, which provide the units for later reasoning instead of treating all images as a flat prompt.

The central reasoning step is cross-image inference. It resolves recurring entities, compares states, infers temporal or spatial relations, discovers links among images, and identifies which observations can ground a claim. The final synthesis stage converts this inferred structure into user-facing artifacts such as summaries, stories, reports, timelines, or memory updates. This expanded view also clarifies the boundary between the VMI pipeline and the agentic framework in the main paper: the pipeline specifies the functional stages needed for collection-level reasoning, whereas

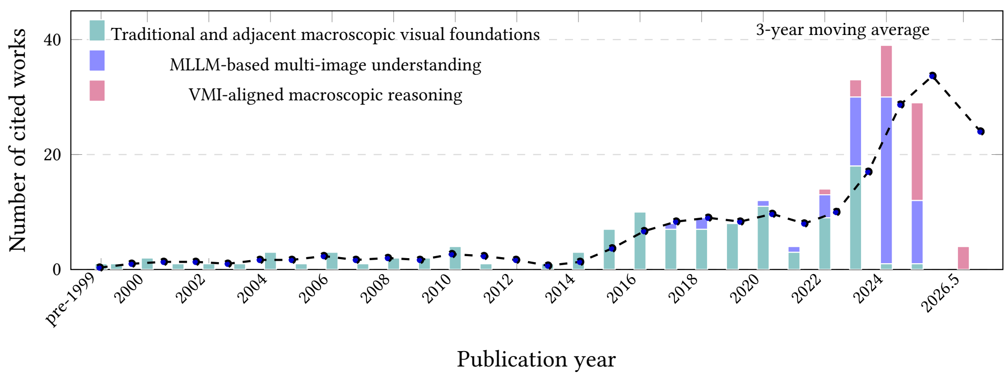

Fig. 5. Publication trends in the cited literature, grouped by dominant role in this survey. Bars show annual stacked counts for traditional and adjacent macroscopic visual foundations, MLLM-based multi-image understanding, and VMI-aligned macroscopic reasoning. Generic agent and tool-use papers are counted as adjacent foundations unless they explicitly construct macroscopic visual evidence. The dashed curve shows a trailing 3-year moving average of total cited works for trend visualization only. The pre-1999 bin contains one classic memory reference, and the 2026.5 bin covers work released up to May 2026.

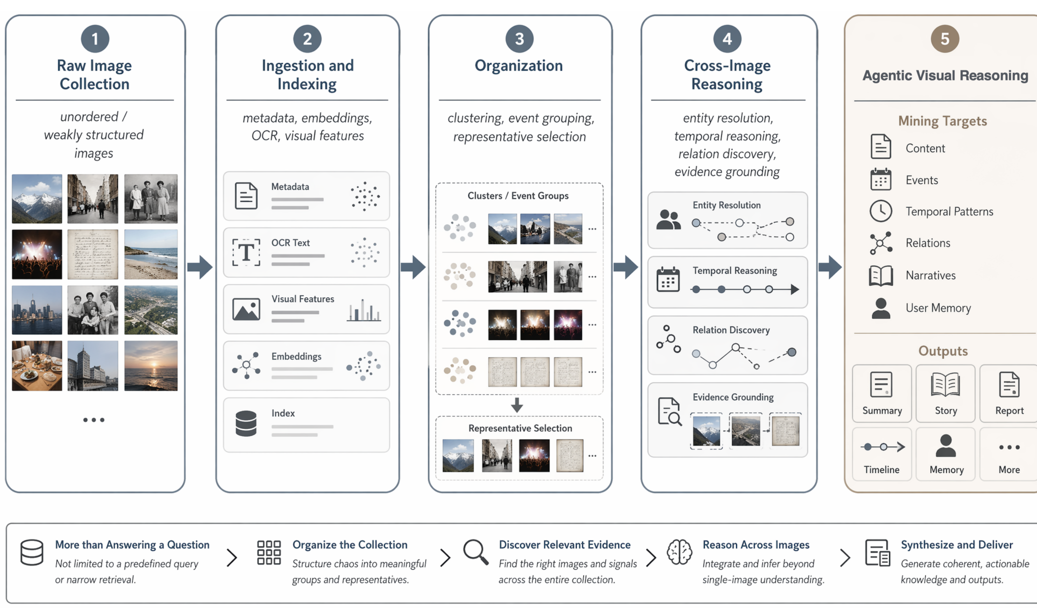

Fig. 6. Detailed pipeline of latent structure induction for VMI. Given an unordered or weakly structured image collection, the system indexes and organizes visual inputs, performs cross-image reasoning to discover entities, events, temporal patterns, relations, and evidence, and finally synthesizes user-facing outputs such as summaries, stories, reports, timelines, or memory.

agentic visual reasoning adds an iterative policy for deciding which stages to invoke, what evidence to inspect next, and when the accumulated evidence is sufficient for delivery.

B.4 Illustrative Example of Evidence-Graph Construction

This section gives a concrete example of how the latent evidence-graph abstraction can be instantiated in a weakly structured image collection. The example is illustrative rather than a proposed benchmark or algorithm; its purpose is to clarify what information may become nodes, edges, attributes, and output claims in a VMI system.

Input context and goal. Consider a personal visual context containing twelve images from a phone album: an airport departure-board photo (I_1), a hotel booking screenshot (I_2), two landmark photos (I_3, I_4), a group photo near the landmark (I_5), a restaurant receipt screenshot (I_6), a metro-map screenshot (I_7), several near-duplicate street photos (I_8, I_9), and three unrelated screenshots or memes (I_{10}, I_{11}, I_{12}). The user goal is to generate a concise travel timeline with cited visual evidence.

Nodes. The evidence graph first contains image nodes for all inputs, but it also introduces higher-level nodes that summarize latent structure. Entity nodes may include `Person-A`, `Landmark-B`, `Hotel-C`, and `Restaurant-D`. Event nodes may include `Arrival`, `Hotel-Stay`, `Landmark-Visit`, and `Dinner`. Claim nodes encode statements that may appear in the final artifact, such as `C1`: "the user traveled to Beijing," `C2`: "the user visited Landmark-B on Oct. 2," and `C3`: "the dinner occurred after the landmark visit."

Edges. Edges encode both organizational structure and evidential support. In this example, representative edges can be written as short graph records:

- I_3 and I_5 have `part-of` edges to `Landmark-Visit`, grouping the landmark images into an event.
- I_4 has a `same-as` edge to `Landmark-B`, linking a visual observation to an entity.
- I_6 has a `supports` edge to `C3`, connecting the receipt to a temporal claim.
- I_{10} has an `unrelated-to` edge to `Hotel-Stay`, recording that an attractive but irrelevant screenshot should not be used as evidence.
- `Landmark-Visit` has a `before` edge to `Dinner`, inferred from timestamps, OCR text, and cross-image context.

Attributes and provenance. Attributes store the evidence behind each node or edge. In this example, they may include:

- `I2.OCR`: "Beijing Hotel, Oct. 1–3";
- `I6.OCR`: "Restaurant-D, 2025-10-02 19:30";
- `C2.support`: $\{I_3, I_4, I_5\}$;
- `C2.provenance`: landmark recognition, timestamp, and the co-occurring group photo;
- `C2.confidence`: 0.91.

The landmark images may additionally store recognition confidence, GPS metadata, or region-level grounding boxes. These attributes distinguish the generated claim from the evidence used to justify it.

Grounded output. A VMI system can then generate a timeline by querying the graph rather than by summarizing all images uniformly:

- Oct. 1: arrival in Beijing, supported by the departure-board photo and hotel booking screenshot (I_1, I_2).
- Oct. 2 afternoon: visit to Landmark-B, supported by landmark and group photos (I_3, I_4, I_5).
- Oct. 2 evening: dinner at Restaurant-D, supported by the receipt screenshot (I_6).

The same graph also explains why unrelated screenshots are excluded and which claims require additional verification. For instance, if landmark recognition is uncertain or the receipt lacks a date,

Table 13. A capability-oriented taxonomy of traditional image sequence and photo collection analysis.

Capability level	Guiding question	Typical output	Modeling cues and representative works
Mining / Indexing	What latent patterns, clusters, or reusable access structures exist in a large image repository?	Visual index; clusters; association patterns; candidate evidence pool	Low-level features, local descriptors, visual vocabularies, similarity search, spatial verification, scalable indexing [22, 43–46, 138, 139]
Selection / Curation	Which images should be retained, highlighted, or removed from a redundant collection?	Representative subset; curated album; slideshow candidate set	Quality, saliency, faces, diversity, redundancy, event coverage, user or social preference [1, 2, 18, 85, 140]
Event Summarization / Organization	How should photos be grouped into meaningful events, episodes, or timelines?	Event clusters; temporal segments; compact album summary	Timestamps, GPS/EXIF, visual similarity, clustering, event recognition, social context, cross-media cues [1, 2, 48, 141]
Temporal Ordering / Storyline Reconstruction	What order or storyline best explains a weakly ordered or jumbled collection?	Ordered sequence; storyline graph; transition structure	Chronology, visual continuity, image-caption alignment, recurrent sequence modeling, graph reconstruction [49, 50, 142]
Narrative / Procedural Abstraction	What high-level story, procedure, or semantic interpretation can be derived from the sequence?	Story; procedural description; narrative or textual summary	Image order, sequential modeling, global-local attention, scene graphs, external knowledge, planning [3, 48, 51, 53, 87, 118, 143–145]
Story Quality / Style / Grounding	How can generated stories be made coherent, expressive, and tied to salient entities?	Rewarded story; sentiment- or style-controlled story; grounded narrative	Coherence rewards, sentiment modeling, topic control, character or entity grounding, scene-graph evidence [53, 145–147]

the agentic reasoning loop may inspect GPS metadata, rerun OCR, compare visual landmarks, or surface uncertainty in the final timeline. This example illustrates the intended role of the evidence graph: it is an intermediate, provenance-aware structure that connects images, entities, events, claims, and support records before producing a grounded artifact.

C Detailed Literature Synthesis and System Settings

C.1 Detailed Traditional Collection-Analysis Tasks

Table 13 summarizes representative traditional task families. Unlike modern MLLM benchmarks, these tasks usually assume that the system is designed for a specific output format, such as a representative subset, an event summary, or a story. Nevertheless, they reveal a recurring theme: image collections must first be structured before they can be meaningfully consumed.

C.1.1 Image Selection and Album Curation. Image selection is one of the earliest forms of collection-level visual analysis. Given a large and redundant photo collection, the system must select a compact subset that preserves important visual content while satisfying constraints such as target length, visual diversity, or user-facing presentation quality. Early systems for automatic image selection considered applications such as slideshows and photobooks, where the selected subset should cover the collection without overwhelming the viewer [1, 140]. Related work on social albums further examined how photo selection can support personal storytelling and public sharing by incorporating user and social context beyond low-level visual quality [2]. Later studies introduced event-specific image importance and album-level event recognition, showing that image importance is partly conditioned on the semantic type and structure of the event [18, 85].

These methods typically rely on signals such as aesthetics, sharpness, face presence, visual saliency, temporal coverage, user preference, and redundancy removal. From the perspective of this survey, their importance lies in treating a photo collection as a structured object rather than a bag of independent images. Selection requires the system to estimate which images are representative, which images are redundant, and which parts of the collection deserve coverage. This makes image selection an early precursor to collection-level organization, evidence selection, and personalized visual memory.

C.1.2 Event Summarization and Album Organization. Album summarization extends image selection by organizing photos into meaningful events or episodes. Personal photo collections are rarely clean sequences: users may take many near-duplicate photos within minutes, omit intermediate moments, mix screenshots with camera photos, or combine multiple events in the same folder. Early personal album summarization and social-album systems therefore combined visual features, timestamps, user context, and social sharing cues to form compact event-level summaries [1, 2]. Later event-aware curation further connected album organization with event recognition and event-specific image importance, suggesting that an album's structure should be interpreted relative to the type of activity it depicts [18]. Traditional summarization methods therefore often exploit weak metadata such as timestamps, GPS coordinates, file names, and social context, together with visual similarity and clustering. The output may be an event-level summary, a timeline, or a set of representative images for each episode.

This line of work is directly relevant to visual macroscopic intelligence because it recognizes that the semantic unit of interest is often not an individual image but an event or sub-collection. In a travel album, for example, meaningful structure may correspond to places visited, activities performed, and people appearing repeatedly. Hierarchical album models further connect event organization with downstream story generation, treating photo selection, event representation, and sequence-level language generation as coupled problems [48]. Related work on joint photo-stream and blog-post summarization shows that album organization can also benefit from weakly aligned textual context, not only from visual similarity [141]. In a personal album, the desired summary may depend on a balance among coverage, diversity, aesthetics, and emotional or social importance. However, traditional album summarization typically remains bounded by predefined objectives and cannot easily support open-ended questions such as what changed over time, which images provide evidence for a claim, or what long-term user preferences can be inferred.

C.1.3 Temporal Ordering and Storyline Reconstruction. A related but distinct family of methods focuses on recovering order or storyline structure from weakly ordered image collections. Unlike album summarization, where the output is often a compact subset or event cluster, ordering methods ask how images should be sequenced so that they form a plausible temporal, causal, or narrative progression. This problem appears in jumbled photo streams, image-caption story datasets, vacation-photo organization, and web community photo collections where the observed upload order may not reflect the underlying event structure.

Representative work reconstructs storyline graphs for image recommendation from web community photos, learns visual storylines with recurrent models that can skip less informative observations, or sorts shuffled images and captions into coherent stories [49, 50, 142]. These studies make chronology and transition structure explicit. For visual macroscopic intelligence, this is important because many later reasoning tasks, such as change analysis, procedural understanding, and memory construction, depend on whether the system can distinguish true temporal progression from arbitrary presentation order.

C.1.4 Narrative Generation and Procedural Abstraction. Beyond selecting, grouping, and ordering images, a more advanced line of traditional research studies how to transform image sequences or photo streams into higher-level semantic descriptions, such as coherent stories, procedural descriptions, or textual summaries. This direction can be viewed as a step from collection organization to collection-level semantic abstraction: the system must decide which visual details are salient, infer relations across images, maintain topical or entity coherence, and produce language that reflects the collection as a whole.

Visual storytelling is a representative task in this direction, though not the only one. The VIST dataset introduced sequential vision-to-language generation by aligning photo streams with both descriptive and story-like language, emphasizing that a story should capture event-level meaning rather than simply concatenate image captions [3]. Procedural resources such as RecipeQA further require systems to understand multi-step cooking processes from image-text recipes [118]. Multi-image summarization similarly targets textual summaries from cohesive image sets, connecting album-level organization with language generation [87]. The analogy to multi-document summarization is useful here: both settings must compress redundant, partially overlapping evidence into a coherent artifact, but large visual contexts additionally require grounding claims to images, regions, and latent visual relations [148–150]. Subsequent methods explored stronger sequence modeling and narrative coherence. Hierarchically attentive RNNs jointly selected summary photos and generated album-level stories [48]; GLAC Net used global-local attention and context cascading to generate stories conditioned on multiple images [143]; Hide-and-Tell explicitly learned to bridge visual gaps in incomplete photo streams [51]; knowledge-enriched storytelling incorporated external knowledge to produce more informative narratives [144]; scene-graph-based methods modeled objects and visual relations before story generation [145]; and question-answer plans connected visual representations with more explicit narrative planning [53].

For this survey, the main significance of narrative generation is that it demonstrates the need for abstraction beyond recognition, event grouping, and ordering. A narrative output must compress a collection into a coherent semantic interpretation, rather than merely list visible objects or select representative images. At the same time, these tasks are usually limited to short, ordered or weakly ordered sequences with fixed output forms. They do not fully address unordered, heterogeneous, or large-scale image collections where the system must first discover what the collection is about and what information is worth extracting.

C.1.5 Story Quality, Style, and Grounding. Beyond generating narrative content, related traditional work also examined whether a visual story is coherent, expressive, controllable, and grounded in salient visual entities. Reward-based work examined what makes a good visual story by combining objectives for relevance, coherence, diversity, and human preference [146]. Sentiment-aware models further introduced affective control, showing that visual stories may need to express emotional trajectories rather than neutral descriptions [147]. Scene-graph and question-answer-plan methods also point toward grounded narrative construction, where objects, relations, and planned questions serve as intermediate structures before text generation [53, 145].

This direction is important because it exposes a limitation of treating visual storytelling as plain sequence-to-text generation. A generated story may be fluent while ignoring key characters, inventing unsupported transitions, or overemphasizing visually salient but narratively irrelevant details. In visual macroscopic intelligence, these concerns become evidence and provenance problems: the system should explain which images, entities, and relations justify a narrative claim, and should distinguish stylistic enrichment from visually grounded facts.

C.2 Detailed MLLM-Based Multi-Image Understanding Literature

C.2.1 From Single-Image VLMs to Multi-Image MLLMs. Early MLLMs and large vision-language models established the general paradigm of coupling visual encoders with language models. Flamingo demonstrated that a visual language model can perform few-shot multimodal learning over interleaved visual and textual inputs [54]. BLIP-2 and InstructBLIP clarified how frozen visual encoders and large language models can be aligned through lightweight querying modules and instruction tuning [55, 151]. LLaVA and LLaVA-1.5 further popularized visual instruction tuning, showing that instruction-following behavior can be extended to image-conditioned dialogue and reasoning through scalable synthetic supervision and improved training recipes [56, 152]. Kosmos-2 and Qwen-VL highlight another foundation for VMI: grounding language responses in visual entities, regions, and text-rich image content rather than treating the image as an opaque context vector [57, 153]. These systems laid the foundation for open-ended visual-language interaction.

However, most early MLLM settings focus primarily on one image or on short visual contexts, whereas recent multi-image datasets and benchmarks explicitly expose the need for inter-image comparison, co-reference, temporal reasoning, and distractor robustness [4, 7, 8, 60]. Multi-image understanding introduces additional sources of difficulty. First, the model must bind entities across images: the same person, object, place, or document may appear under different viewpoints, scales, or lighting conditions, and recent correspondence studies show that even strong MLLMs often fail at this matching problem [72, 154]. Second, the model must determine which images are relevant to a query and which are distractors. Third, it must reason over relations that are not visible in any single image, such as change, order, co-reference, contradiction, or multi-hop evidence [65–67]. These requirements make multi-image understanding more than a simple extension of single-image visual question answering.

C.2.2 Multi-Image Instruction Tuning. A key development in the MLLM era is multi-image instruction tuning. Instead of relying on a model's implicit ability to process multiple visual inputs, recent work constructs training data and architectures that explicitly expose models to interleaved image-text contexts, image sequences, and multi-image questions. MANTIS constructs interleaved multi-image instruction data and targets capabilities such as co-reference, reasoning, comparison, and temporal understanding [4]. LLaVA-NeXT-Interleave extends large multimodal models to multi-image, video, and 3D scenarios, showing that interleaved visual inputs can provide a unified interface for heterogeneous visual contexts [5]. mPLUG-Owl3 further emphasizes long image-sequence understanding, targeting scenarios where many related images must be processed jointly [59]. MMDU combines a multi-turn multi-image dialogue benchmark with instruction-tuning data, highlighting the importance of conversational multi-image understanding [60]. Related work on interleaved vision-language instruction following further shows that explicit mixed image-text instruction formats can improve a model's ability to maintain image identity and follow multi-step multimodal tasks [155]. Qwen2-VL improves high-resolution and video-oriented perception through dynamic visual tokenization and multimodal position encoding [39]. In parallel, large interleaved training collections such as OBELICS, OmniCorpus, and Multimodal C4 provide document-like training contexts in which multiple images and text spans are naturally interwoven [61, 156, 157].

These works mark an important departure from traditional pipelines. Rather than designing a fixed algorithm for selecting, grouping, or narrating images, multi-image instruction tuning trains a general model to follow natural-language tasks over multiple visual inputs. This added flexibility comes with new dependencies: performance is strongly influenced by instruction-data diversity, input order, visual token budget, and the model's ability to preserve image identity across a long context.

Table 14. Representative multi-image MLLM systems and training frameworks.

System	Year	Input setting	Core mechanism	VMI relevance	Limitation
MANTIS [4]	2024	Interleaved multi-image instruction data.	Multi-image instruction tuning for co-reference, comparison, reasoning, and dialogue.	Establishes a training recipe for query-driven cross-image reasoning.	Small or preselected contexts; no collection-level structure.
LLaVA-NeXT-Interleave [5]	2024	Interleaved image-text, video, and 3D inputs.	Unified interleaved instruction tuning over heterogeneous visual contexts.	Provides a general interface for multi-image and mixed visual inputs.	Does not explicitly model evidence graphs or provenance.
mPLUG-Owl3 [59]	2024	Long image sequences and interleaved contexts.	MLLM design targeting long image-sequence understanding.	Moves multi-image modeling toward longer visual contexts.	Still mostly answer- or dialogue-oriented.
Qwen2-VL [39]	2024	High-resolution images, videos, and multi-image inputs.	Dynamic visual tokenization and multimodal position encoding.	Strengthens perception and OCR-like visual evidence extraction.	Broad VLM; not designed for latent collection organization.
LLaVA-OneVision [158]	2024	Single-image, multi-image, and video tasks.	Unified visual task transfer across input formats.	Bridges image, video, and multi-image instruction following.	Limited explicit evidence attribution.
LongLLaVA [38]	2024	Long visual contexts up to many images.	Hybrid architecture for efficient scaling to large image sets.	Addresses the context-window bottleneck for VMI.	Long input does not guarantee structure discovery.
LongVILA / V2PE [159, 160]	2024	Long videos or long multimodal contexts.	Scaling and variable visual position encoding for long-context VLMs.	Improves retention of visual identity and position over long contexts.	Primarily context modeling, not evidence-state reasoning.
MAmmoTH-VL / Insight-V [74, 75]	2025	Reasoning-intensive multimodal instructions.	Rationale-rich instruction data and long-chain visual reasoning.	Improves explicit reasoning over visual evidence.	Reasoning traces may remain weakly grounded to images.
MiCo [76]	2025	Multi-image contrastive reasoning tasks.	Reinforcement learning over contrasting image sets.	Targets fine-grained cross-image comparison and cue linking.	Task-specific post-training; limited open-ended VMI coverage.

Table 14 summarizes representative multi-image MLLM systems at the method level. The table complements the capability-oriented taxonomy by making explicit the input regime, core mechanism, VMI relevance, and remaining limitation of each system family.

C.2.3 Multi-Image Question Answering and Reasoning. Multi-image question answering has become the dominant evaluation format for MLLM-based multi-image understanding. In this setting, the model receives several images and a natural-language question, and it must aggregate evidence across the images to produce an answer. Depending on the task, the required reasoning may involve comparison, counting, co-reference, visual search, temporal ordering, state transformation, or multi-hop inference. Earlier resources such as NLVR and Spot-the-Diff already exposed core forms of cross-image reasoning, including relational language grounding and difference description over similar image pairs [122, 123].

Recent benchmarks reveal that current MLLMs still struggle with these capabilities. General evaluation suites such as MME, MMBench, SEED-Bench, and BLINK show that strong models can answer many language-conditioned visual tasks while still missing fine-grained perception, spatial relations, or visually necessary evidence [6, 165–168]. Hallucination-focused evaluations such as POPE and HallusionBench further show why answer fluency is insufficient: models may confidently assert objects, relations, or visual premises that are not supported by the image [169, 170]. MIRB evaluates perception, knowledge, reasoning, and multi-hop reasoning across multiple images

Table 15. Representative visual RAG and agentic visual reasoning systems.

System	Year	Visual context	Mechanism	VMI relevance	Limitation
Visual ChatGPT / MM-ReAct [106, 161]	2023	Mostly single images or short visual tasks.	LLM controller routes requests to visual foundation models and tools.	Early template for visual tool orchestration.	Not designed for persistent large visual contexts.
HuggingGPT [114]	2023	General multimodal AI tasks.	LLM plans tasks and assigns them to external expert models.	Shows modular tool selection and task decomposition.	Generic orchestration without visual evidence provenance.
VisProg / ViperGPT [107, 108]	2023	Single images or small visual inputs.	Generates executable programs over visual modules.	Makes intermediate reasoning operations explicit and inspectable.	Program traces are local; weak collection-level state.
Chameleon [115]	2023	Multimodal reasoning tasks.	Plug-and-play compositional reasoning with external tools.	Useful pattern for assembling VMI perception and reasoning modules.	Not specialized for image collections or provenance.
AVIS [116]	2023	Visual information seeking tasks.	Autonomous agent chooses external resources and iteratively refines evidence.	Closest early pattern for active visual evidence search.	Search setting is narrower than macroscopic collection analysis.
Interactive retrieval / MRA [83, 162]	2024–2025	Image retrieval with text or composed queries.	Query rewriting, reasoning, and agentic retrieval for image search.	Provides goal-conditioned retrieval primitives for VMI.	Returns ranked images, not structured artifacts.
VisRAG [12]	2024	Multimodal document pages.	VLM-based page retrieval and generation without OCR-first parsing.	Page-level visual evidence retrieval and grounded generation.	Mostly document QA; limited open-ended structure induction.
ViDoRAG [100]	2025	Large visual document corpora.	Hybrid visual-text retrieval plus multi-agent coarse-to-fine reasoning.	Strong example of agentic visual-document evidence search.	Slide/document heavy; LLM-judge evaluation.
DeepImageSearch [84]	2026	Personal visual histories.	Agentic context-aware retrieval over long photo histories.	Moves retrieval toward collection-scale visual evidence exploration.	Retrieval-only output; little synthesis or memory evaluation.
ReMe / XSkill [110, 117]	2025–2026	Agent experience and skills, sometimes multimodal.	Procedural memory refinement and continual skill learning.	Suggests memory mechanisms for cumulative VMI agents.	Memory is not yet tied to full visual provenance.

[62], while ReMI targets reasoning problems whose evidence is distributed across image sets [63]. MuirBench evaluates robust multi-image understanding across diverse task types and multi-image relations, emphasizing distractors and unanswerable variants [7]. MMIU provides a broad evaluation suite covering multiple image relationships and many task categories, including spatial and 3D-oriented tasks related to ScanQA and SQA3D [8, 126, 127]. MMRB further expands this direction with structured multi-image reasoning tasks and chain-of-thought-style annotations [64]. Broader MLLM benchmarks such as MMMU, MM-Vet, and MathVista stress integrated visual reasoning, expert knowledge, and mathematical or diagrammatic reasoning, while Mementos and MIBench more directly test image-sequence and multiple-image reasoning [65, 124, 125, 171, 172]. Collectively, these benchmarks indicate that multi-image reasoning requires more than language-level reasoning over image captions; it requires faithful visual evidence aggregation and relation modeling.

C.2.4 Temporal, Transformation, and Long-Context Reasoning. Several recent studies show that multi-image reasoning is not only about aggregating evidence from a small set of images, but also about modeling order, change, and long-range visual context. Temporal image-sequence

Table 16. Domain-specific large visual contexts under the visual macroscopic intelligence framework. Specialized domains differ in macroscopic structure, evidence requirements, and risk profiles.

Domain	Typical macroscopic structure	Evidence and verification requirement	Key challenge
Personal visual histories	Mixed photos, screenshots, receipts, social images, metadata, and user history.	Time/location/entity links; user feedback; provenance for memories.	Privacy, identity errors, editable memory.
Medical image sequences	Multi-view, multi-modal, or longitudinal scans with clinical metadata and reports.	Slice-, region-, view-, or time-point-level evidence; expert verification; calibrated uncertainty [77].	High stakes, scarce labels, modality shifts.
Remote-sensing time series	Observations of the same region across time, sensors, seasons, or resolutions.	Before-after evidence; geo-spatial alignment; change regions; time/sensor metadata [95, 163].	Clouds/seasonality, scale, false alarms.
Scientific charts and visual documents	Figures, charts, tables, formulas, scanned pages, slides, and document images.	OCR/layout; axes, legends, visual marks, and page provenance [12, 98, 99, 129, 164].	Visual-text-numeric alignment, wrong-page attribution.
Product and inspection collections	Product galleries, multi-view objects, defect images, batch records, and specification screenshots.	Viewpoint correspondence; local defect regions; product attributes and specification evidence.	Near duplicates, subtle variants, quality criteria.

benchmarks such as Mementos and TempVS/Burn After Reading test whether models can capture event order rather than relying on language priors or dataset shortcuts [65, 66]. Transformation-oriented resources ask models to infer how a visual state changes across images. Earlier state-to-transformation reasoning exposed this problem in controlled settings, while VisualTrans extends it to real-world human-object interaction scenarios involving spatial, procedural, and temporal changes [67, 86].

Long-context work addresses a different bottleneck: even when all relevant images are available, the model may lose image identity, overlook sparse evidence, or fail to attend to a small number of critical frames. MileBench, MMNeedle, and Visual Haystacks evaluate long multimodal contexts and needle-in-a-haystack retrieval over visual inputs [40, 41, 173]. LongLLaVA and LongVILA explore model scaling for long visual contexts, while V2PE improves multimodal long-context capability through variable visual position encoding [38, 159, 160]. These directions are important for visual macroscopic intelligence because real collections are rarely limited to a few curated images; relevant evidence may be sparse, distant, or embedded in document-like image streams.

C.2.5 Grounding, Comparison, and Correspondence Across Images. Beyond answering a final question, a visual macroscopic intelligence system should identify which visual evidence supports its answer. This motivates grounding, comparison, and correspondence across images. Grounding asks the model to associate a response with relevant images or regions. Comparison asks it to identify similarities, differences, or changes. Correspondence asks it to match entities, scenes, or visual cues across inputs. These capabilities are essential for collection-level reasoning because real image collections often contain multiple plausible but competing pieces of evidence.

This problem has an important pre-MLLM lineage in multi-view geometry and robotic perception. Feature matching, epipolar constraints, structure-from-motion, SLAM, and dense reconstruction explicitly model which observations across images refer to the same 3D point, object surface, or camera trajectory [34–37, 44, 174]. VMI does not replace these geometric tools; rather, it extends the question from “which pixels or views correspond?” to “which visual observations support a semantic claim, event, or user-facing artifact?” This distinction is especially important for embodied, robotic, multi-view, and spatial-reasoning corpora, where semantic evidence should remain consistent with pose, visibility, occlusion, and view overlap.

Existing multi-image benchmarks increasingly include such relational requirements. For example, MuirBench explicitly considers diverse multi-image relations and robustness to misleading or insufficient visual evidence [7]; MMIU covers multiple categories of image relationships and evaluates models across a wide range of tasks [8]. Recent grounding-oriented efforts further move beyond answer accuracy: PRIMA studies multi-image pixel-grounded reasoning segmentation [69], MC-Bench evaluates multi-context visual grounding in which models must localize target instances across multiple images [70], Migician targets free-form multi-image grounding [128], and GeM-VG extends this direction toward generalized multi-image visual grounding [71]. Visual correspondence studies such as MMVM and VLM2-Bench further show that even strong MLLMs can fail at identifying whether visual instances correspond across images or whether explicit visual cues should be linked across inputs [72, 154]. These evaluation efforts suggest a shift from treating multiple images as independent context items to treating them as a relational and evidential structure. Nevertheless, fine-grained evidence localization remains underdeveloped compared with final-answer accuracy, and models may still hallucinate cross-image links that are not visually supported.

C.2.6 Reasoning Post-Training and Domain-Specific Evaluation. Beyond building broader benchmarks, recent work increasingly attempts to improve multi-image reasoning through richer supervision, explicit reasoning traces, and reinforcement learning. MAmmoTH-VL constructs large-scale multimodal instruction data with intermediate rationales to elicit reasoning-intensive behavior [74], while Insight-V studies long-chain visual reasoning with MLLMs [75]. MiCo explores multi-image contrastive reinforcement learning, encouraging models to link fine-grained visual cues across similar and different images [76]. Related reinforcement-learning efforts for multi-image grounding indicate that post-training objectives may be especially useful when final-answer supervision is too weak to teach correspondence and evidence localization.

Another underrepresented direction is domain-specific multi-image evaluation. Medical image sequences require models to compare views, modalities, and time points while grounding findings to the correct slice or region; MedSG-Bench targets this setting with medical image sequence grounding tasks [77]. Multi-visual mathematical reasoning is evaluated by MV-MATH, where several visual contexts may jointly define the problem [175]. Visual quality comparison benchmarks such as VQualA further test whether MLLMs can compare single images, pairs, and multi-image groups using low-level and perceptual quality criteria [176]. These domain and quality-oriented tasks broaden multi-image understanding beyond generic QA, showing that different corpora require different notions of evidence, comparison, and correctness.

C.3 Representative Visual RAG and Agentic Visual Reasoning Systems

Table 15 summarizes representative systems that operationalize retrieval, tool use, programmatic reasoning, visual information seeking, or memory. Unlike the benchmark table, this table focuses on system mechanisms and clarifies which components can be reused for VMI-style evidence search and artifact generation.

C.4 Domain-Specific Macroscopic Visual Settings

Domain-specific large visual contexts should not be viewed merely as applications of generic multi-image reasoning. They impose different assumptions about macroscopic structure, evidence sufficiency, risk, and acceptable failure modes. Under an evidence-graph view, these domains differ in what counts as a valid node, edge, and support set. A medical claim may require slice-level or region-level support, a remote-sensing claim may require geo-registered before-after evidence, and a chart-based claim may require linking visual marks to OCR text, axes, numerical scales, and page provenance.

Table 16 compares representative domain-specific large visual contexts under the VMI framework. The purpose is not to provide exhaustive domain surveys, but to show why a single generic multi-image benchmark is insufficient: each domain changes the evidence schema, verification standard, and cost of failure.

This domain view complements the general VMI pipeline. Personal albums emphasize privacy and memory correction; medical collections emphasize grounded and conservative evidence; remote sensing emphasizes geo-temporal alignment and change support; and scientific documents emphasize layout- and chart-aware provenance. A unified VMI system should therefore preserve a common abstraction over entities, events, claims, and evidence, while allowing domain-specific evidence schemas and verification rules.

D Evaluation Resources and Automatic Metrics

Existing evaluation resources can be viewed as proxy tests for individual VMI capabilities, including multi-image QA, temporal reasoning, grounding, long-context retrieval, visual-document RAG, and visual-history search. This section separates the resource landscape from metric operationalization: Section D.1 summarizes representative datasets and benchmarks, while Section D.2 describes how VMI-style outputs can be scored once structures, evidence, and traces are exposed.

D.1 Evaluation Resource Landscape

Table 17 summarizes representative recent benchmarks at the dataset level. The table makes explicit each benchmark's input scale, task format, annotation signal, metric, VMI capability, and limitation, while emphasizing that no single resource constitutes a full evaluation of visual macroscopic intelligence.

D.2 Operationalizing Automatic Metrics

The evaluation dimensions in the main paper should not be interpreted as human-only rubrics. A practical benchmark can require each system to produce a structured trace

$$\hat{T} = (\hat{Z}, \{\hat{E}_c\}_{c\in\text{Claims}(\hat{Y})}, \hat{\tau}),$$

where $\hat{Z}$ is the predicted macroscopic structure, $\hat{E}_c$ is the cited evidence set for claim c, and $\hat{\tau}$ records retrieval, tool-use, verification, and memory operations. Once reference or partially annotated structures Z^*, evidence sets E^*_c, and target artifacts are available, many metrics become automatically computable. For example, structure induction can be scored by graph edge precision/recall or clustering ARI/NMI; evidence faithfulness can be scored by citation precision and recall,

$$\text{CitationP} = \frac{\sum_c |\hat{E}_c \cap E^*_c|}{\sum_c |\hat{E}_c|}, \qquad \text{CitationR} = \frac{\sum_c |\hat{E}_c \cap E^*_c|}{\sum_c |E^*_c|},$$

and region-level support can be evaluated by IoU or pointing accuracy. Table 18 summarizes an annotation-to-metric path. Learned visual verifiers or LLM judges may be used as auxiliary scorers for open-ended artifacts, but the benchmark should make image IDs, regions, graph edges, and process traces first-class outputs so that deterministic or verifier-assisted automatic scoring is possible.

Table 17. Representative benchmark and dataset attributes for multi-image and VMI-related evaluation.

Benchmark	Year	# images / context size	Task	Annotation	Metric	VMI capability	Limitation
BLINK [6]	2024	3,807 MCQs; single or multiple images.	Fine-grained visual perception over 14 tasks.	Multiple-choice answers with task-specific visual prompts.	Accuracy.	Perception and spatial/correspondence primitives.	Mostly short contexts; not collection discovery.
MIRB [62]	2024	925 samples; 2–42 images, 3.78 on average.	Perception, knowledge, reasoning, multi-hop QA.	MCQ/free-form labels from curated, synthetic, or existing sources.	Accuracy.	Cross-image comparison and multi-hop reasoning.	Small curated sets; limited provenance.
ReMI [63]	2024	13 tasks; 2–6 images per problem.	Multi-image reasoning in math, charts, code, maps, physics, and logic.	Programmatic labels and manually checked answers.	Accuracy; relaxed numeric accuracy; ERP.	Heterogeneous evidence integration.	Controlled domains; little real collection noise.
MuirBench [7]	2025	11k images; 2.6k MCQs.	Robust multi-image QA.	Answerable and unanswerable variants.	Accuracy.	Distractor and relation robustness.	Final-answer focus; weak traces.
MMIU [8]	2025	77,659 images; 11,698 MCQs; 52 tasks.	Multi-image understanding across 7 relationships and 5 modalities.	Metadata-derived MCQs via rules or GPT-4o.	Accuracy.	Broad relational and modality coverage.	Large but still query-driven.
MMDU [60]	2024	110 dialogues; 2–20 images; over 1,600 questions; up to 18k tokens.	Multi-turn multi-image dialogue.	GPT-4o-assisted questions with human-revised long answers.	GPT-4o judge; average accuracy / overall score.	Conversational memory over multi-image context.	Small benchmark split; judge-dependent scoring.
Mementos [65]	2024	4,761 image sequences; many 4–14 frames, some over 15.	Generate episode descriptions from image sequences.	Human descriptions with object and behavior keywords.	Object/behavior precision, recall, F1.	Temporal event and behavior reasoning.	Ordered sequences rather than unordered collections.
MMNeedle [40]	2025	40k images, 560k captions, 280k pairs; up to 640 subimages.	Locate caption needles in long visual haystacks.	COCO captions plus generated positive/negative locations.	Existence, index, exact accuracy.	Long-context visual evidence retrieval.	Synthetic placement; limited semantic synthesis.
Visual Haystacks [41]	2025	COCO-derived haystacks; single-needle or 2–3-needle challenges; systems tested beyond 100 images.	Vision-centric needle finding and multi-image QA.	COCO-derived object queries and binary-style answers.	Accuracy / task success.	Sparse visual evidence search.	Focuses retrieval stress, not artifact generation.
PRIMA / M4Seg [69]	2024	About 744k reasoning-segmentation QA pairs.	Multi-image reasoning segmentation.	QA pairs with pixel-level masks.	Segmentation IoU and answer accuracy.	Region-level evidence grounding.	Segmentation-centric; limited large-context organization.
MC-Bench [70]	2025	About 2k samples; image pairs.	Multi-context visual grounding from open-ended prompts.	Manually annotated instance-level targets.	Grounding accuracy / IoU-style localization.	Cross-image region grounding.	Pairwise contexts only.
VisRAG [12]	2024	Six document QA sources; page images as retrieval units; top-1–3 pages for generation.	Vision-based document RAG.	VQA-derived query-page pairs plus synthetic training data.	MRR@10, Recall@10, relaxed answer accuracy.	Page-level evidence retrieval and grounded generation.	Mostly document/page QA.
ViDoRAG / ViDoSeek [100]	2025	About 1.2k questions over ~6k slide/page images.	Large-corpus visual document retrieval-reason-answer.	Expert-built or refined queries with reference pages and answers.	Recall, MRR@5, GPT-4o-judged accuracy.	Agentic visual-document evidence search.	Slide-heavy corpus; LLM-judge dependence.
DeepImageSearch [26]	2026	57 histories; 109k photos; 122 queries.	Context-aware retrieval over visual histories.	Human-verified query–target sets.	Exact match, F1; ranking metrics.	Visual-history evidence retrieval.	Retrieval only; little synthesis.

Table 18. An annotation-to-metric path for automatically evaluating visual macroscopic intelligence.

Dimension	Required benchmark signal	Automatically computable metrics
Coverage	Important entities, events, evidence subsets, or weighted slots.	Weighted entity/event recall; rare-evidence recall; representative coverage; duplicate ratio.
Structure induction	Reference or partial entity links, event groups, timelines, and graph edges.	Cluster purity, ARI/NMI; entity-linking F1; graph edge precision/recall/F1; temporal order violations.
Cross-image reasoning	Pairwise or multi-hop relation labels, change labels, ordering labels.	Relation accuracy/F1; pairwise consistency; Kendall's τ; change-detection F1.
Evidence faithfulness	Claim-level gold evidence IDs, regions, OCR spans, or contradiction labels.	Citation precision/recall; region IoU; unsupported-claim rate; contradiction rate; abstention accuracy.
Retrieval and organization	Gold relevant images, ranked evidence pools, cluster labels.	Recall@k, MRR, nDCG; clustering purity; redundancy reduction; retrieval cost per supported claim.
Synthesis utility	Required output schema, reference timelines/reports, task-specific slots.	Slot completeness; factual precision/recall against references; artifact-validity checks; calibrated uncertainty.
Agentic process quality	Logged actions, tool calls, inspected images, verification outcomes, memory edits.	Success under budget; tool-call cost; evidence discovery rate; verification pass rate; memory update precision; privacy-leak rate.